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FINAL PROJECT REPORT 2025/2026

OPTIMISED DESIGN OF A BATTERY ENERGY STORAGE SYSTEM FOR MINIMISING ELECTRICITY COSTS IN PHOTOVOLTAIC- SUPPLIED RESIDENTIAL HOMES

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Word Count: 7789

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Abstract

This study focuses on the modelling and simulation of a residential photovoltaic PV system integrated with battery energy storage BESS and grid interaction. The main objective is to develop an energy management strategy capable of efficiently coordinating power flow between the PV source, battery, and grid under varying operating conditions. With the increasing demand for renewable energy and the need to reduce grid dependency, PV-BESS systems have become a key solution for residential energy management, as highlighted in recent literature.

The system was developed using MATLAB/Simulink, where a PV array, DC-DC boost converter with Maximum Power Point Tracking MPPT, grid-connected inverter, and battery model with state-of-charge SOC monitoring were implemented. An energy management control strategy was designed based on PV generation, load demand, and SOC thresholds. The system was then evaluated under multiple operating scenarios representing different irradiance and load conditions.

The results demonstrate that the system maintains a stable supply to the load across all conditions while effectively adapting to variations in PV generation. The MPPT controller enabled efficient power extraction, and the energy management strategy prioritised local energy utilisation while reducing reliance on the grid. The findings highlight the practical potential of integrated PV-BESS systems for residential applications, with opportunities for further improvements in control strategies and system optimisation.

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List of Abbreviations and Mathematical Symbols

PV – Photovoltaic

BESS – Battery Energy Storage System

SOC – State of Charge

MPPT – Maximum Power Point Tracking

EMS – Energy Management System

DC – Direct Current

AC – Alternating Current

PLL – Phase-Locked Loop

PWM – Pulse Width Modulation

PI – Proportional-Integral Controller

LCL – Inductor–Capacitor–Inductor Filter

THD– Total Harmonic Distortion

UK– United Kingdom

Acknowledgements

I would like to sincerely thank my supervisor, Dr. Mahdi Salimi, for their guidance, patience, and continuous support throughout this project. Their advice and feedback really helped me stay on track, especially during the more challenging stages of this work.

Finally, I want to thank my family and friends for always being there for me. Their support, encouragement, and belief in me made a big difference, and I truly couldn't have done this without them.

CHAPTER ONE

INTRODUCTION

1.1 Background of Project

The increasing penetration of residential PV systems in the United Kingdom has been driven by rising electricity prices, government decarbonisation targets, and growing consumer awareness of sustainable energy solutions. While rooftop solar installations enable households to generate renewable electricity, a fundamental limitation remains: PV generation is intermittent and does not naturally align with residential load demand profiles. Peak solar production typically occurs during daytime hours, whereas household demand often peaks during evening periods when solar output is minimal. Figure 1 illustrates the typical daily mismatch between photovoltaic generation and household load demand.

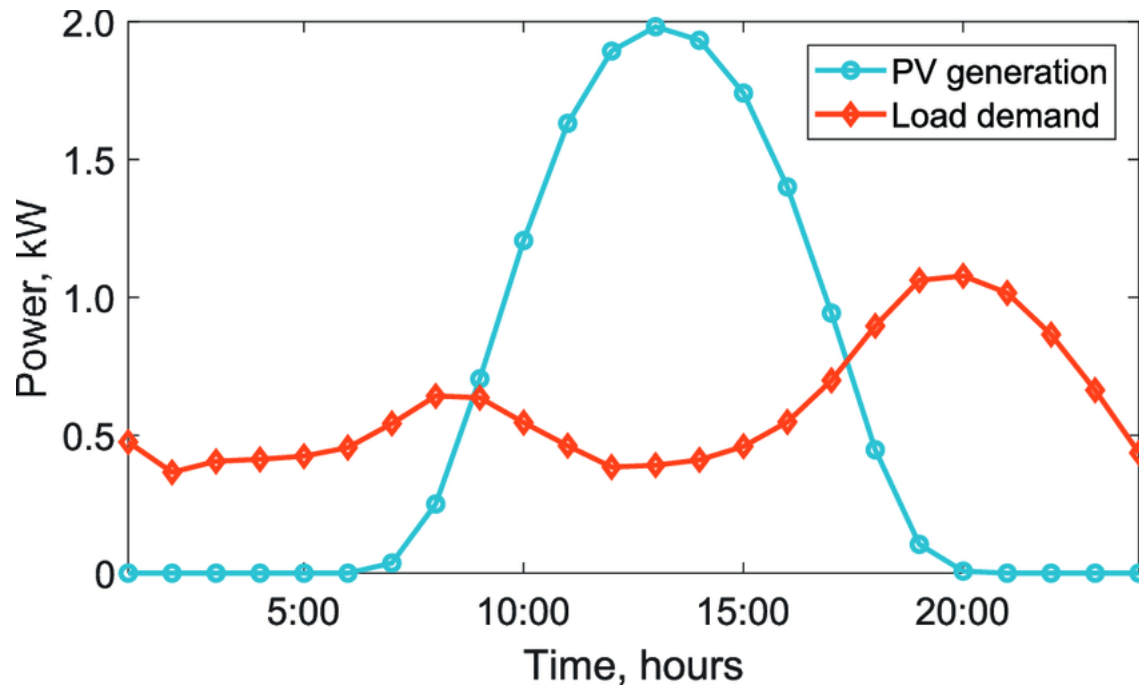


Figure 1 Typical daily mismatch between PV generation and residential load demand (Sharma, Haque and Aziz, 2019).

This temporal mismatch leads to two inefficiencies. First, excess PV generation during daylight hours is exported to the grid at relatively low feed-in tariffs. Second, electricity must be imported at higher retail tariffs during the evening, even in households equipped with solar panels. As a result, the economic potential of residential PV systems is not fully realised without additional energy management strategies.

BESS have emerged as a promising solution to address this mismatch. By storing surplus PV energy during periods of high generation and discharging during periods of high demand, batteries can reduce grid dependency and potentially lower electricity bills. However, the financial viability of residential battery systems depends strongly on operational control strategies, SOC management, tariff structures, and system sizing.

Poorly optimised battery dispatch may result in minimal economic benefit despite significant capital investment. A typical residential photovoltaic system integrated with battery energy storage is illustrated in Figure 2.

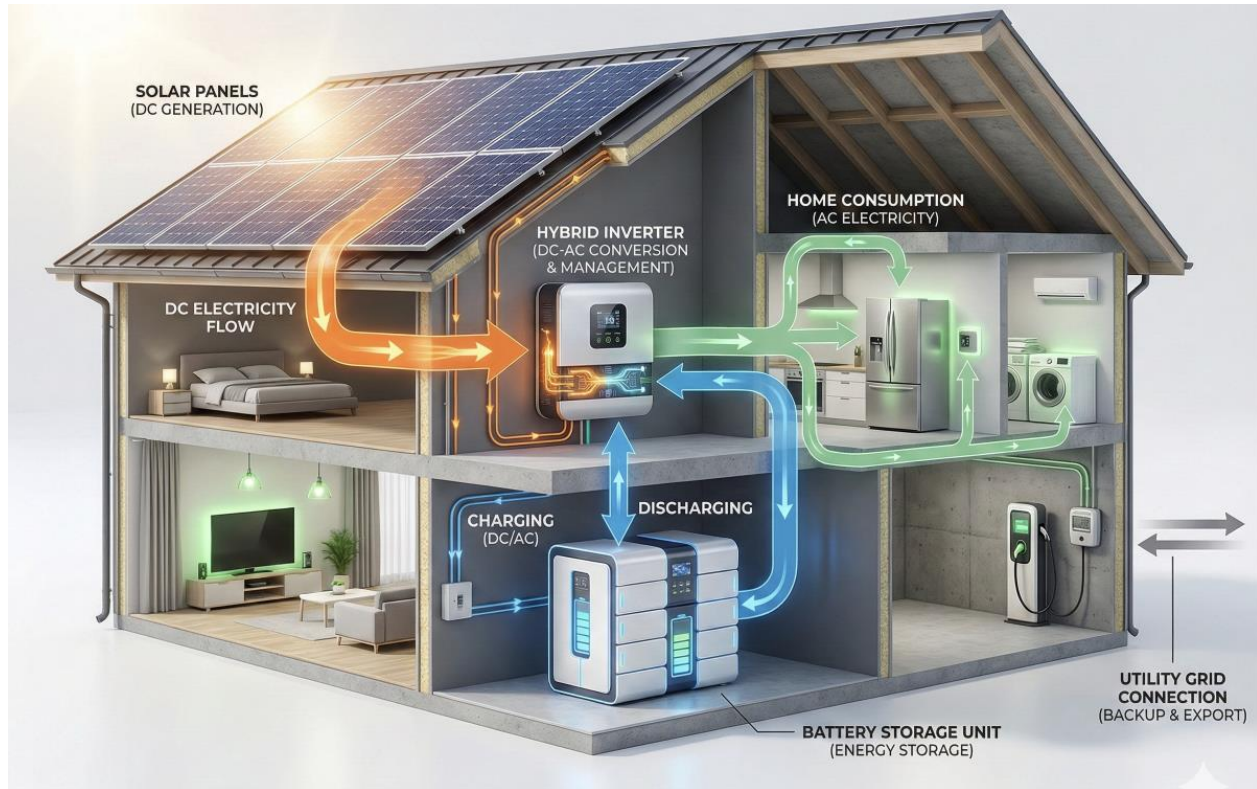


Figure 2 Conceptual illustration of a residential photovoltaic system integrated with battery energy storage and grid connection (Cornett, 2025).

Recent research has explored various approaches to PV-battery integration, including rule-based energy management systems, model predictive control strategies, and tariff-optimised dispatch algorithms. Nevertheless, many studies either focus purely on technical modelling without economic evaluation, or perform financial analysis without detailed system-level simulation of power electronics and control logic. There remains a

need for an integrated modelling framework that combines electrical system simulation, control implementation, and economic assessment under realistic UK residential tariff conditions.

This project therefore investigates the modelling and optimisation of a grid-connected PV-BESS residential system using MATLAB/Simulink. The work integrates power electronics design, MPPT, SOC control, and tariff-based cost evaluation to assess both technical performance and economic impact.

1.2 Aims and Objectives of the Project

1.2.1 Project Aims

The aim of this project is to design and evaluate a photovoltaic-battery-grid integrated system in order to minimise residential electricity costs through optimised energy management and battery operation.

1.2.2 Project Objectives

- 1 To develop a detailed simulation model of a residential photovoltaic system using MATLAB/Simulink.
- 2 To design and implement a DC-DC boost converter with MPPT control.
- 3 To model a grid-connected inverter with appropriate filtering for AC power injection.
- 4 To implement a battery model with SOC monitoring and operational constraints.
- 5 To develop an energy management control strategy based on PV generation, load demand, and SOC thresholds.

- 6 To simulate multiple operating scenarios representing different irradiance and load conditions.

1.3 Definition of relevant terms

Photovoltaic (PV): A technology that converts solar irradiance directly into electrical energy using semiconductor materials.

Battery Energy Storage System (BESS): An electrochemical storage system used to store electrical energy for later use.

State of Charge (SOC): The percentage of available battery capacity relative to its maximum rated capacity.

Maximum Power Point Tracking (MPPT): A control technique used to maximise the power output of a photovoltaic system by continuously adjusting operating voltage.

Energy Management System (EMS): A control framework responsible for determining power flow between PV, battery, load, and grid.

DC-link: A common DC bus that connects different power electronic stages and enables energy exchange between the PV system, battery, and inverter.

Phase-Locked Loop (PLL): A control system used to synchronise the inverter output with the grid voltage by tracking its phase and frequency.

Pulse Width Modulation (PWM): A switching technique used to control power electronic devices by varying the duty cycle of the switching signal.

1.4 Scope of the Project

This project focuses on the simulation-based modelling and evaluation of a single-phase residential PV-battery-grid system. The system is analysed under representative operating conditions using computational tools.

The scope includes:

- Design and modelling of power electronic converters.
- Implementation of MPPT control using a perturb-and-observe algorithm.
- SOC-based rule-driven battery energy management.
- Simulation of multiple irradiance and load scenarios.

The scope excludes:

- Hardware implementation and experimental validation.
- Battery degradation modelling and lifecycle analysis.
- Advanced grid stability studies.
- Time-of-use tariff optimisation.
- Thermal modelling of power electronic components.

These exclusions ensure that the study remains focused on system-level operational optimisation and economic performance assessment within a defined modelling framework.

1.5 Report Outline

Chapter 1 Introduces the background, aims, objectives, and scope of the study.

Chapter 2 Presents a review of relevant literature on residential PV systems, battery storage integration, control strategies, and economic modelling approaches.

Chapter 3 Describes the research methodology, including system modelling, converter design, control implementation.

Chapter 4 Presents the simulation results and discusses system performance under various operating conditions.

Chapter 5 Summarises the key findings, draws conclusions, and outlines recommendations for future work.

CHAPTER TWO

LITERATURE REVIEW

2.1 Integration of Battery Energy Storage in Grid-Connected PV Systems

The rapid deployment of PV systems in residential and commercial environments has introduced several operational challenges for low-voltage distribution networks. High levels of PV penetration may result in voltage rise, reverse power flow, and increased intermittency due to the variable nature of solar energy. BESS have therefore emerged as an effective solution for improving grid stability and enhancing the utilisation of locally generated renewable energy.

Deshmukh et al. (2023) investigate a grid-connected PV system integrated with battery storage through a bidirectional DC–DC converter and inverter stage. Their simulation-based study demonstrates that battery integration can significantly improve system stability and increase the utilisation of solar generation under varying irradiation conditions. However, the study mainly focuses on system modelling under standard operating conditions and provides limited techno-economic analysis of the proposed architecture.

Similarly, Pannawan et al. (2024) design and experimentally validate a 3-kW single-phase grid-connected battery inverter using an LCL-filtered voltage source converter. Their results demonstrate effective DC-link regulation and harmonic mitigation, achieving a THD below 1.5%. While the study highlights the importance of inverter

control and filtering in PV-battery systems, the focus is primarily on power electronics implementation rather than system-level optimisation for residential energy management.

Mariano and Urbanetz (2022) explore the integration of PV and battery storage within a university microgrid environment. Their case study demonstrates that energy storage can reduce peak demand and improve load management within the system. However, the analysis is largely oriented towards commercial-scale applications and microgrid management rather than residential-level optimisation.

To better understand how batteries are integrated into residential PV systems, several architectures of PV-battery hybrid inverters have been proposed.

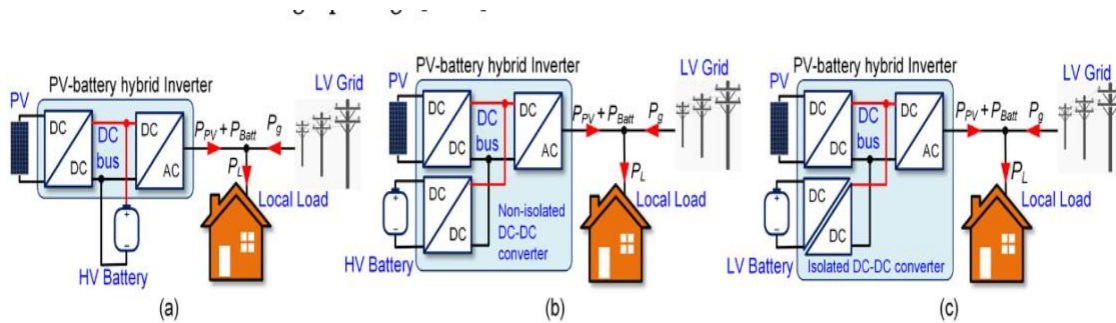


Figure 3 Typical architectures of PV-battery hybrid inverter systems showing different battery integration approaches and bidirectional power exchange between PV generation, battery storage, local loads, and the grid (Source: Pannawan et al., 2024).

Collectively, these studies confirm that PV-BESS architectures provide important technical benefits, including improved grid interaction, increased renewable energy

utilisation, and enhanced operational flexibility. Nevertheless, many existing studies concentrate on hardware implementation or large-scale microgrid systems.

Consequently, there remains a need for modelling approaches that analyse both system dynamics and economic performance within residential PV-battery applications.

2.2 Energy Management Strategies for PV-Battery Systems

The energy management strategy applied to a PV-battery system plays a critical role in determining its economic performance. Various control strategies have been proposed in the literature, ranging from simple rule-based approaches to advanced optimisation and artificial intelligence techniques.

Li and Preindl (2023) compare several EMS approaches, including load levelling, peak shifting, and model predictive control for grid-connected PV-battery microgrids. Their results indicate that MPC achieves superior cost optimisation when accurate forecasting data are available. However, the computational complexity of MPC and its reliance on reliable predictions may limit its practical application in small-scale residential systems. To manage the interaction between PV generation, battery storage and load demand, an energy management system is typically implemented.

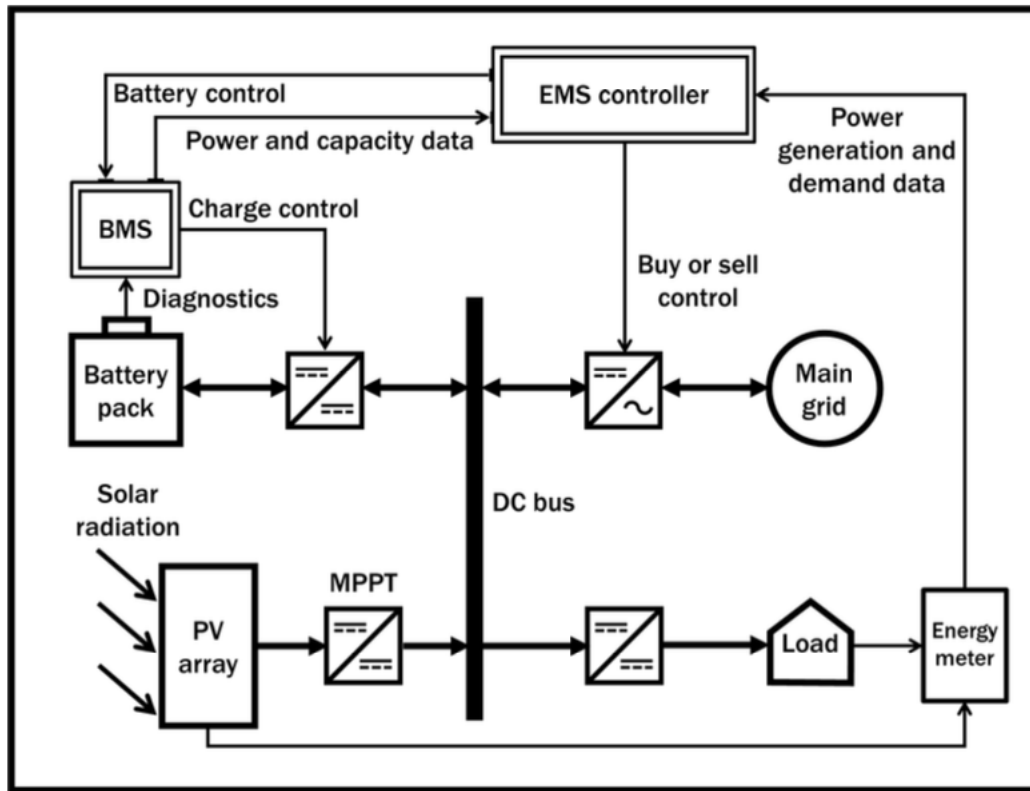


Figure 4 Microgrid system diagram illustrating the energy flow between PV generation, battery storage, the load and the grid under the supervision of an energy management system (Source: Li & Preindl, 2023).

Norouzi et al. (2025) investigate the influence of electricity pricing structures - including net metering, feed-in tariffs, time-of-use, and real-time pricing - on PV-battery system optimisation. Their analysis shows that tariff structures significantly affect the economic viability of battery storage, often having a greater impact on profitability than purely technical system parameters. Nevertheless, their optimisation approach relies on genetic

algorithms and assumes detailed forecasting capabilities that may not always be available in real residential settings.

Han et al. (2022) conduct a techno-economic optimisation study of PV-battery systems applied to residential clusters. Their sensitivity analysis demonstrates that the economic feasibility of battery systems is highly dependent on electricity prices, solar irradiation levels, and household load demand. Importantly, their findings suggest that maximising self-consumption alone does not necessarily guarantee financial profitability.

Song et al. (2024) provide a comprehensive review of mathematical programming and AI-based optimisation methods used for BESS scheduling. Their work concludes that advanced optimisation techniques can significantly improve system performance and operational efficiency. However, such approaches may introduce considerable computational burden and implementation complexity.

More recently, Real et al. (2024) implement deep reinforcement learning for residential PV-battery energy management. Their proposed approach achieves electricity bill reductions of up to 35% compared to conventional optimisation strategies. Despite these promising results, AI-based control methods require large training datasets and significant computational resources, which may limit their widespread adoption in residential applications.

Overall, the literature demonstrates that advanced optimisation techniques can significantly improve PV-battery system performance. However, for residential-scale systems there exists a clear trade-off between optimisation performance and practical

implementation. In many cases, simpler rule-based or simulation-based strategies may offer a more practical and robust solution for residential energy management.

2.3 Power Quality and LCL Filter Design

Grid-connected PV-battery inverters must comply with strict power quality standards in order to operate safely within electrical distribution networks. Harmonic mitigation is therefore an important aspect of inverter design, and LCL filters are widely used to reduce switching harmonics and improve power quality.

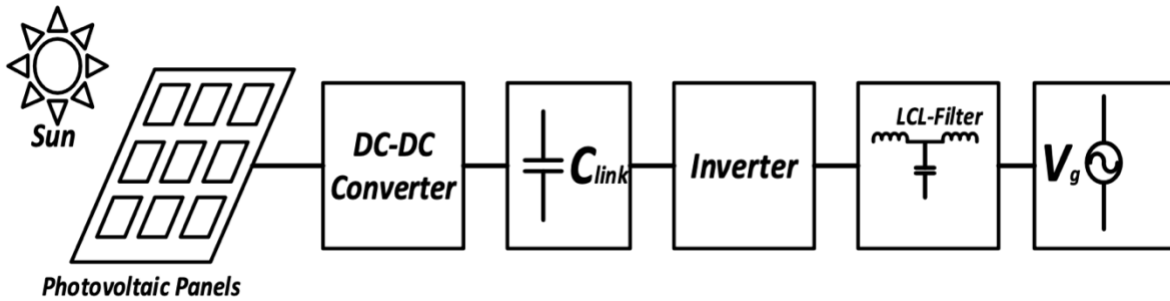


Figure 5 Structure of a grid-connected photovoltaic system incorporating a DC-DC converter, inverter and LCL filter for harmonic mitigation before grid connection (Source: Adamas-Pérez et al., 2024).

Adamas-Pérez et al. (2024) propose an optimised design methodology for LCL filters in single-phase grid-connected microinverters. Their approach demonstrates that appropriate analytical sizing of filter components can reduce passive component size while maintaining effective harmonic attenuation performance. This contributes to improved system efficiency and reduced physical footprint.

Pannawan et al. (2024) further demonstrate the importance of coordinated control between the inverter and the LCL filter in maintaining low harmonic distortion and stable DC-bus regulation. Their experimental results confirm that LCL filtering is essential for ensuring grid compliance in residential PV-battery systems.

Although these studies provide valuable insights into inverter design and harmonic mitigation, many focus primarily on hardware implementation without integrating filter performance into broader system-level techno-economic models. As a result, the interaction between power quality design and system-level economic optimisation remains an area requiring further investigation.

2.4 Techno-Economic Optimisation and Financial Viability

The economic viability of PV-battery systems is influenced by a combination of technical design parameters, electricity pricing structures, and policy incentives. Several studies have attempted to optimise system design and operation in order to maximise economic benefits.

Laksahapsoro et al. (2025) apply mixed-integer linear programming to optimise the design and operation of PV-BESS systems for commercial buildings in the United Kingdom. Their results indicate a potential payback period of approximately 5.5 years under favourable electricity pricing conditions. However, the study focuses on commercial applications and assumes the availability of advanced optimisation tools.

Han et al. (2022) demonstrate that battery profitability is expected to increase over time as battery costs decline and energy policies evolve. Nevertheless, their analysis highlights the sensitivity of investment decisions to market conditions and electricity price variations.

Similarly, Norouzi et al. (2025) show that real-time pricing mechanisms can significantly improve self-consumption ratios in PV-battery systems. However, these pricing schemes may also increase investment payback periods unless supportive policy incentives or subsidies are available.

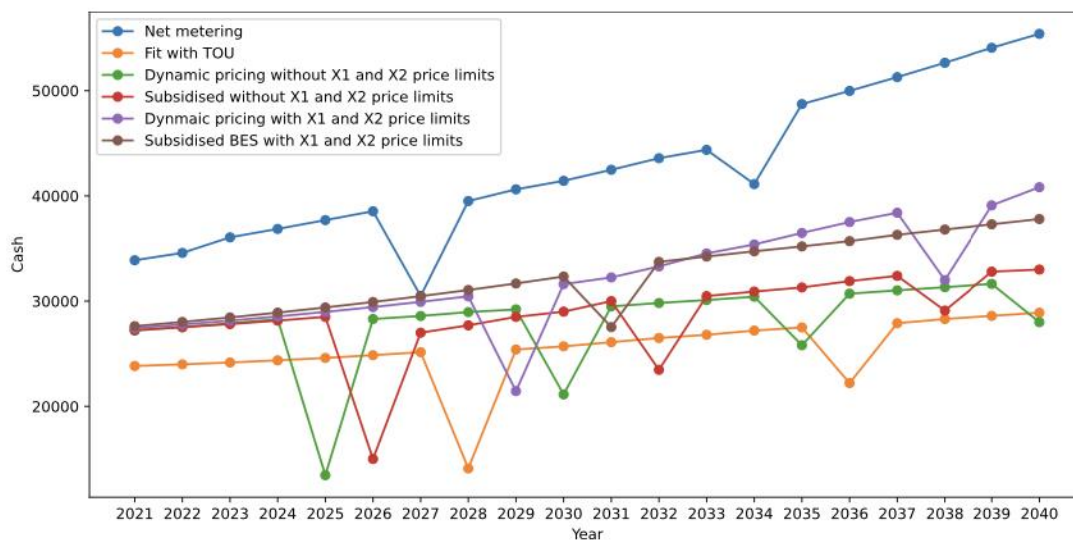


Figure 6 Cash-flow comparison of PV-battery systems under different electricity pricing policies over the project lifetime (Source: Norouzi et al., 2025).

As shown in Figure 6, electricity tariff structures significantly influence the financial performance of PV-battery systems. Net metering scenarios typically produce higher cumulative financial returns compared to feed-in tariff or time-of-use pricing schemes.

Overall, existing research indicates that tariff structures and electricity pricing mechanisms exert a significant influence on the return on investment of battery energy storage systems. However, relatively limited literature integrates residential-scale tariff assumptions with dynamic simulation-based modelling of PV-battery systems. This gap highlights the need for modelling approaches that simultaneously analyse technical system behaviour and economic performance in residential energy applications.

CHAPTER THREE

METHODOLOGY

3.1 Research Methodology

This project adopts a computational modelling and simulation-based approach to evaluate the performance of a residential grid-connected photovoltaic system integrated with a Battery Energy Storage System. The objective of this approach is to examine how battery storage can enhance solar energy utilisation, improve power flow control, and reduce reliance on grid electricity within a residential context.

The study is centred on the development of a detailed model in MATLAB/Simulink. The model integrates photovoltaic generation, battery storage, power electronic converters, and a rule-based control strategy within a unified simulation framework. This enables the system behaviour to be analysed under a range of operating conditions without the need for a physical prototype, thereby reducing cost and allowing greater flexibility in testing and evaluation.

The proposed system consists of a PV–battery configuration designed to store excess solar energy and supply it when photovoltaic generation is insufficient. Its performance is assessed through the analysis of key parameters, including power flow dynamics, battery state of charge, DC-link stability, and grid power exchange across multiple scenarios. In this way, the methodology provides a structured and practical approach for determining

whether the integration of battery storage can improve both the technical performance and economic viability of residential PV systems.

3.2 Materials

This study was carried out using a simulation-based approach implemented in MATLAB/Simulink R2023b with the Simscape Electrical library. Standard Simulink blocks were used to model the main system components, including the photovoltaic array, battery storage system, DC–DC converters, DC-link capacitor, single-phase inverter, LCL filter, and grid interface. In addition, dedicated control subsystems were implemented for both the inverter and the battery, including PWM-based inverter control and a battery charge/discharge control strategy governed by SOC thresholds.

The electrical simulation environment was configured using the powergui block in discrete mode to accurately represent the switching behaviour of the power electronic converters. The model was executed using a discrete solver, with a power stage sampling time of 1 μs and a control system sampling time of 10 μs .

3.3 Methods

The complete model was developed by combining a set of interconnected subsystems representing the main parts of the PV–battery energy system. These subsystems include the generation stage, storage stage, power conversion interface, control loops, and energy management logic. Each part was first modelled individually and then integrated

into a single grid-connected architecture. The following subsections describe the main elements of the model.

3.3.1 System Architecture

Figure 7 shows the overall structure of the proposed grid-connected photovoltaic–battery energy storage system. The system is built around a common DC bus, which acts as the main point where energy is shared between the photovoltaic source, the battery, and the grid interface.

The photovoltaic array provides the main source of energy by converting solar irradiance into electrical power. This energy is delivered to the DC bus, where it can be used within the system depending on the operating conditions. The battery storage system is also connected to the same DC bus, allowing it to store excess energy when available and supply power when needed.

The DC bus is further linked to a single-phase grid inverter, which enables the system to exchange energy with the utility grid. This connection allows the system to either export surplus energy or import power when local generation and storage are not sufficient.

Overall, this architecture provides a simple and flexible way to manage energy flow between generation, storage, and the grid. It allows the system to adapt to different operating conditions while maintaining stable and coordinated operation.

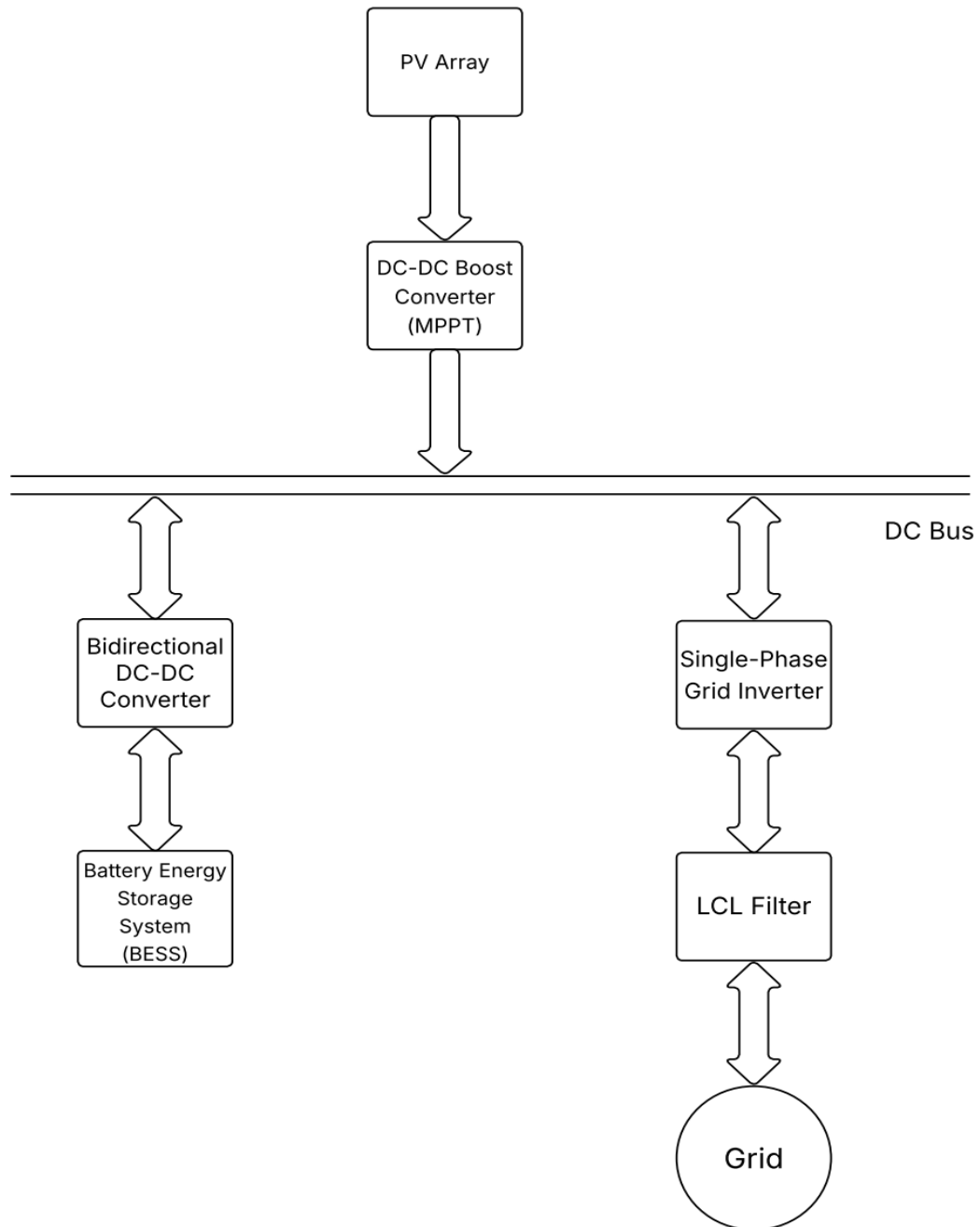


Figure 7 Block diagram of the proposed grid-connected photovoltaic–battery energy storage system.

3.3.2 MPPT Control Using Perturb and Observe Algorithm

The Maximum Power Point Tracking technique is implemented using the Perturb and Observe algorithm in order to maximise the power extracted from the PV. Due to the nonlinear characteristics of PV systems, the operating point varies with environmental conditions such as irradiance and temperature. Therefore, an MPPT controller is required to continuously adjust the operating point and ensure that the PV system operates at its maximum power point.

The P&O algorithm operates by measuring the PV voltage and current and calculating the instantaneous power. The controller then evaluates the variation in power (dP) and voltage (dV) between consecutive sampling instants. Based on the sign of these variations, the algorithm determines whether the operating point is moving towards or away from the maximum power point.

If a perturbation in voltage leads to an increase in power, the algorithm continues to adjust the reference in the same direction. Conversely, if the power decreases, the direction of perturbation is reversed. In this work, the algorithm generates a reference DC-link voltage ($V_{dc,ref}$) which is used to regulate the operating point of the PV system. This iterative process enables effective tracking of the maximum power point under varying operating conditions.

The operational logic of the implemented P&O MPPT algorithm is illustrated in Figure 8.

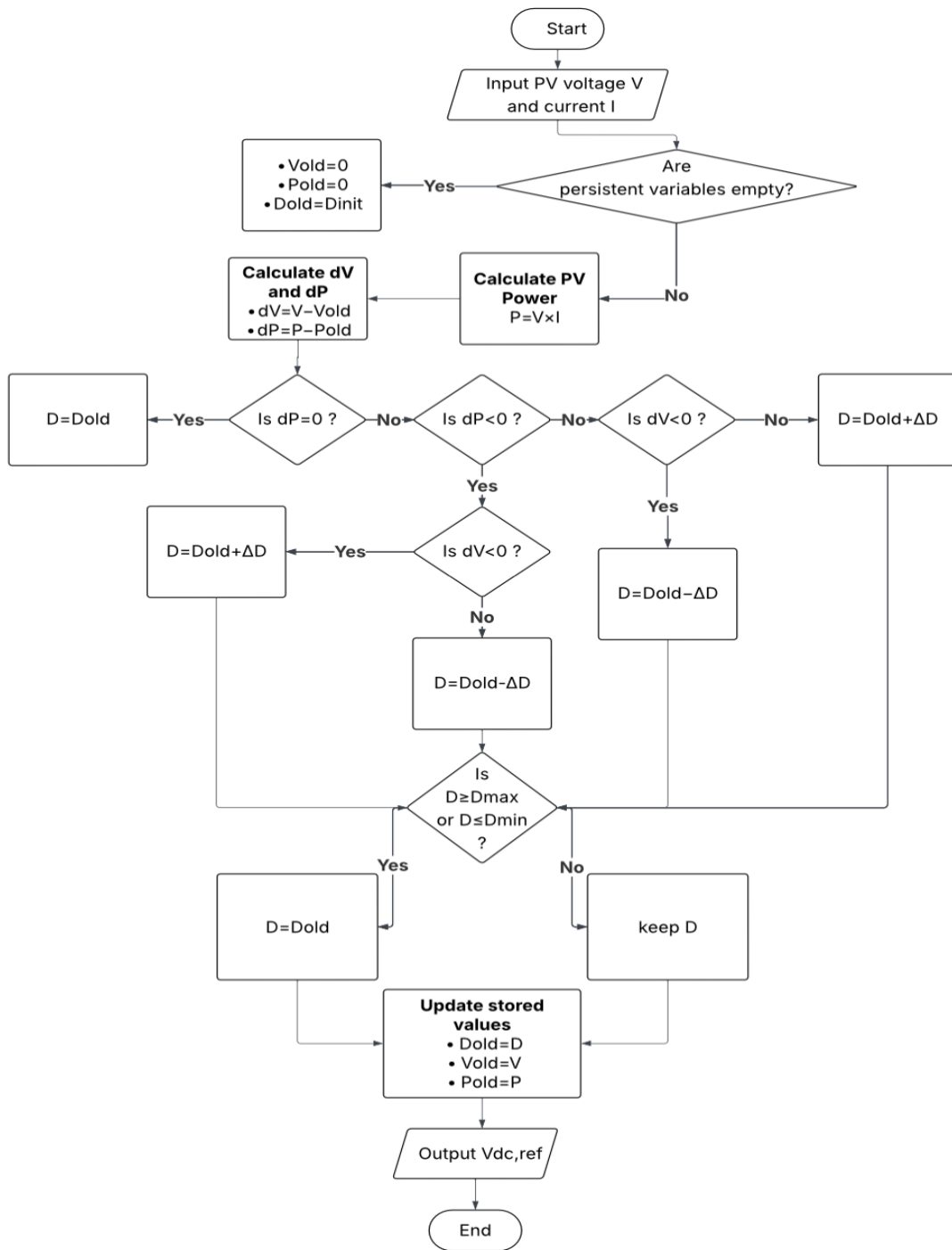


Figure 8 Flowchart of the Perturb and Observe MPPT algorithm used to generate the DC-link voltage reference ($V_{dc,ref}$).

As shown in Figure 8, the algorithm begins by acquiring the PV voltage and current, followed by the computation of the instantaneous power. The changes in power and voltage are then determined and evaluated through a sequence of decision blocks. These decision stages dictate whether the reference DC-link voltage should be increased or decreased by a predefined perturbation step. To ensure stable operation, the reference value is constrained within predefined limits. Finally, the current values are stored for use in the next iteration of the algorithm.

In addition to the algorithmic representation, the practical implementation of the P&O MPPT controller in Simulink is shown in Figure 9.

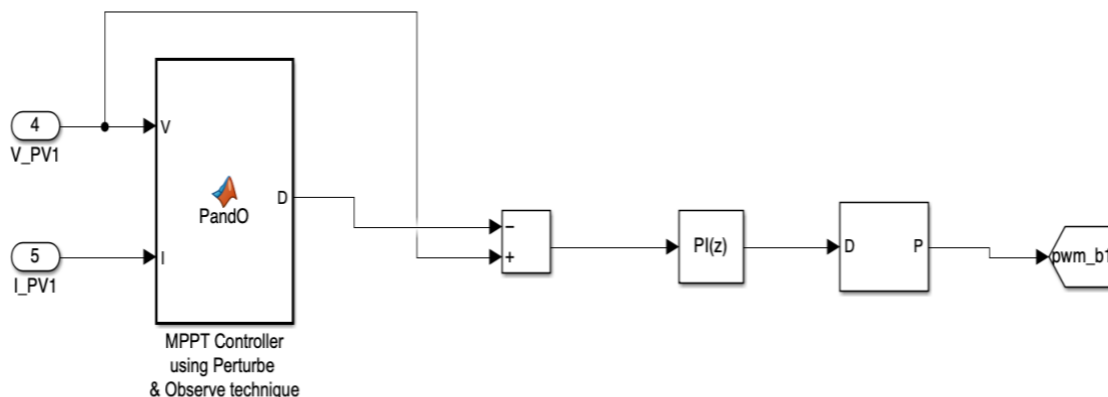


Figure 9 Simulink implementation of the P&O MPPT controller integrated with a PI regulator and PWM generation.

As illustrated in Figure 9, the PV voltage and current are used as inputs to the MPPT controller, which computes the reference DC-link voltage based on the P&O algorithm. This reference is then compared with the measured DC-link voltage, and the resulting

error signal is processed by a discrete PI controller. The PI controller generates a control signal that regulates the DC-DC converter. A PWM generator subsequently converts this signal into switching pulses, ensuring proper operation of the converter, and enabling the PV system to operate close to its maximum power point.

Although the P&O method is simple and widely adopted, it introduces small steady-state oscillations around the maximum power point due to its perturbation nature. However, its low computational complexity and robustness under varying irradiance conditions make it suitable for this application.

3.3.3 Power Electronics Interface

The power conversion stage forms the electrical interface between the photovoltaic generator, the battery energy storage system, and the single-phase grid. Its purpose is to regulate the power flow between these subsystems while maintaining stable DC and AC operating conditions. Similar modelling frameworks have been adopted in previous studies to evaluate system performance under different operating conditions without the need for physical implementation (Deshmukh et al., 2023).

On the photovoltaic side, the PV array is connected to a **boost DC–DC converter** that operates under the control of the MPPT algorithm. This converter adjusts the operating point of the PV array by regulating the switching duty cycle, allowing the system to extract the maximum available power from the solar panels. The boosted output is transferred to the **DC-link**, which acts as the central energy bus of the system.

The DC-link is stabilised using a **DC-link capacitor**, which reduces voltage ripple and decouples the dynamic behaviour of the PV converter, the battery converter, and the inverter stage. Maintaining a stable DC-link voltage is essential for reliable inverter operation and consistent power transfer to the grid.

Energy storage is integrated through a bidirectional DC–DC converter connected to the DC bus and the battery pack. This converter enables controlled charging and discharging of the battery by regulating the battery current. During periods of excess photovoltaic generation, the converter operates in charging mode, transferring energy from the DC bus to the battery. When solar generation is insufficient, the converter operates in the reverse direction, allowing the battery to discharge and support the system.

The DC-link supplies power to a single-phase full-bridge (H-bridge) inverter, which converts the DC voltage into AC power compatible with the grid. The inverter switching signals are generated through a PWM modulation scheme implemented in the control system.

To ensure power quality and limit switching harmonics, the inverter output is connected to the grid through an LCL filter. This filter attenuates high-frequency components generated by inverter switching and ensures that the injected current meets grid quality requirements before reaching the point of common coupling.

Together, these components establish the main energy conversion pathway of the system, enabling controlled power transfer between the photovoltaic array, the battery storage system, and the utility grid. The complete MATLAB/Simulink implementation of

the grid-connected photovoltaic system with battery energy storage, including the PV array, DC–DC converters, DC-link, inverter, and grid interface, is illustrated in Figure 10.

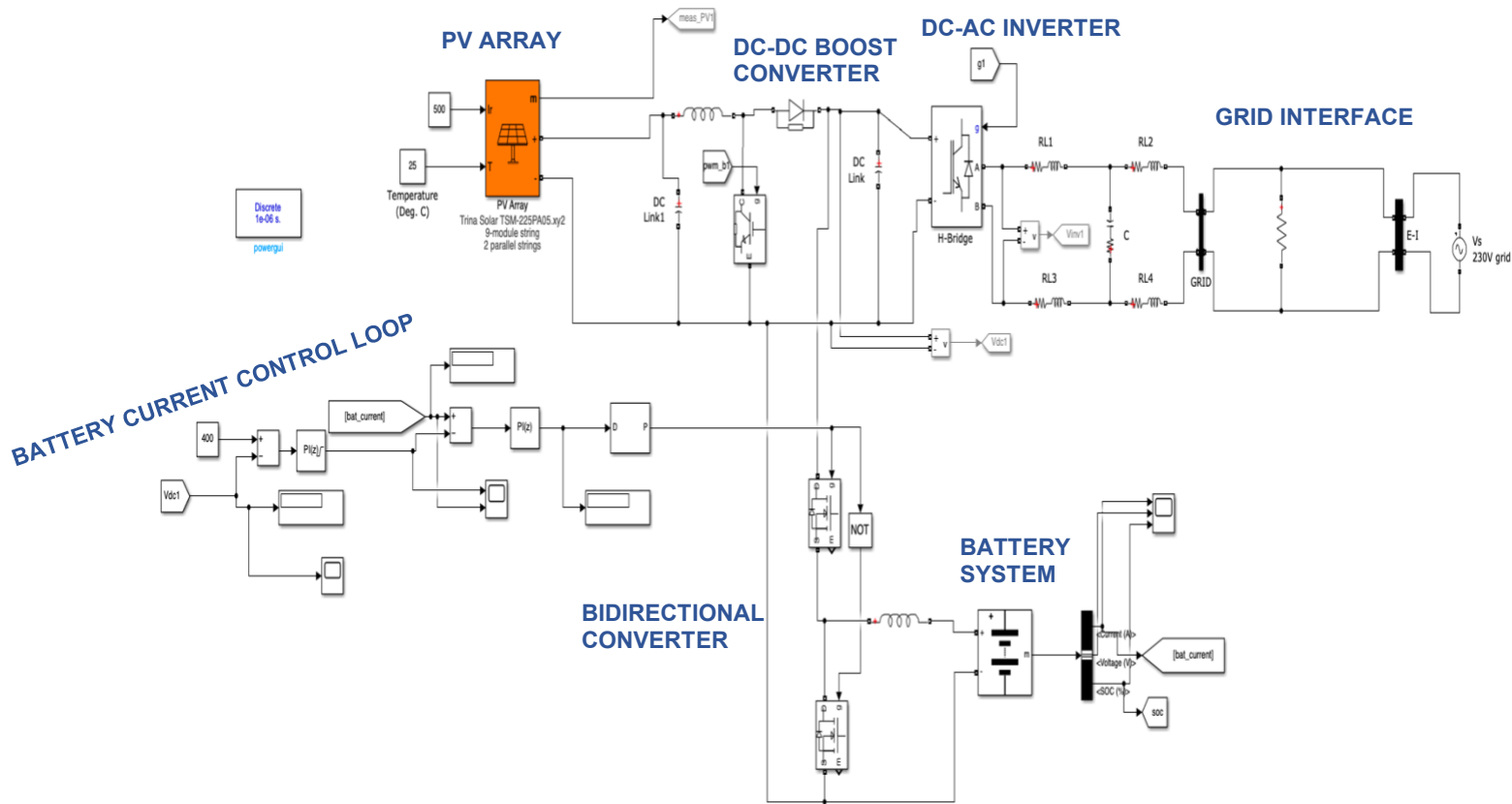


Figure 10 MATLAB/Simulink implementation of the proposed grid-connected PV–battery energy storage system.

3.3.4 Control Strategy

The grid-connected inverter's operation is controlled through a hierarchical control structure, whose principal aim is to maintain stable power conversion and precise current injection to the grid. The grid-connected inverter's control system is comprised of four

major functional blocks: grid synchronization and measurement, current regulation, reference voltage generation, and PWM switching control. The overall inverter control system's architecture is illustrated in Figure 11.

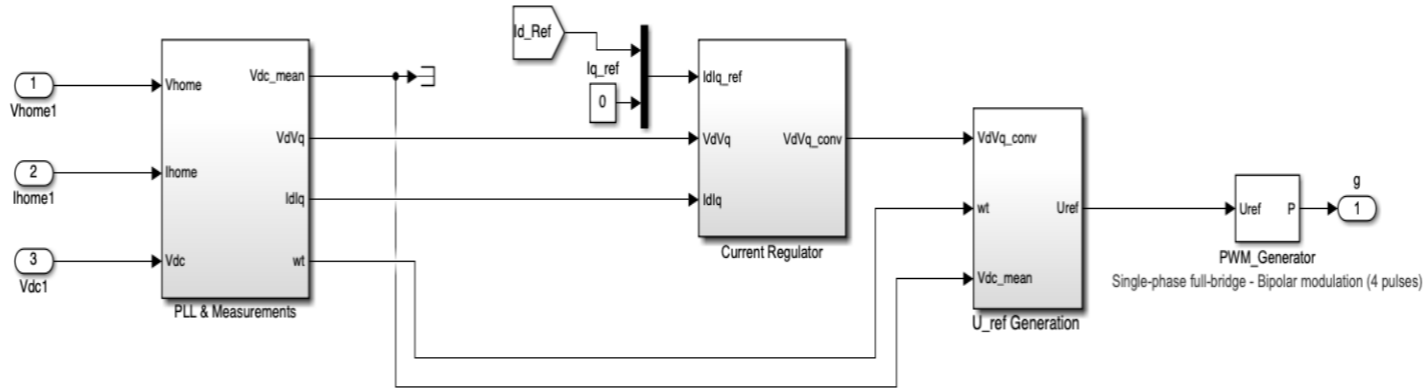


Figure 11 Overall inverter control architecture.

The first block in the grid-connected inverter's control system is responsible for grid synchronization and measurement. A PLL subsystem is employed to synchronize the grid-connected inverter with the grid voltage. The PLL extracts the grid's angular frequency and phase angle, which are necessary to transform the measured signals to the synchronous reference frame. The grid voltage, inverter current, and DC voltage are measured and processed in this subsystem. A single-phase dq transformation is employed to transform the measured signals to the dq frame, which simplifies the current regulation process and allows precise current regulation.

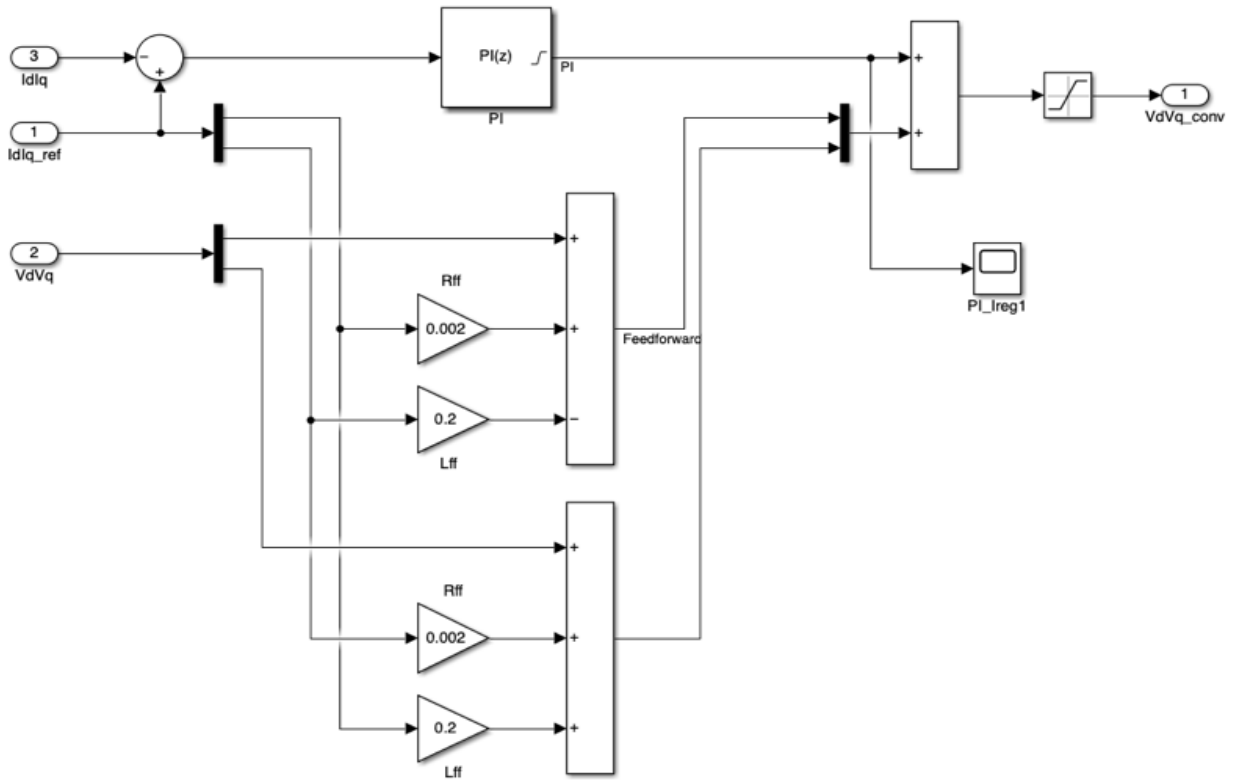


Figure 13 Simulink implementation of the inverter current regulator with feedforward compensation.

The third stage of the control system is responsible for the generation of the reference voltage signal for the inverter. The voltage reference signal from the current regulator and the grid angle from the PLL block are combined to generate a sinusoidal reference signal that is synchronized with the voltage in the grid. The reference signal generated by the control system is the desired voltage that the inverter needs to produce in order to generate the desired current in the grid.

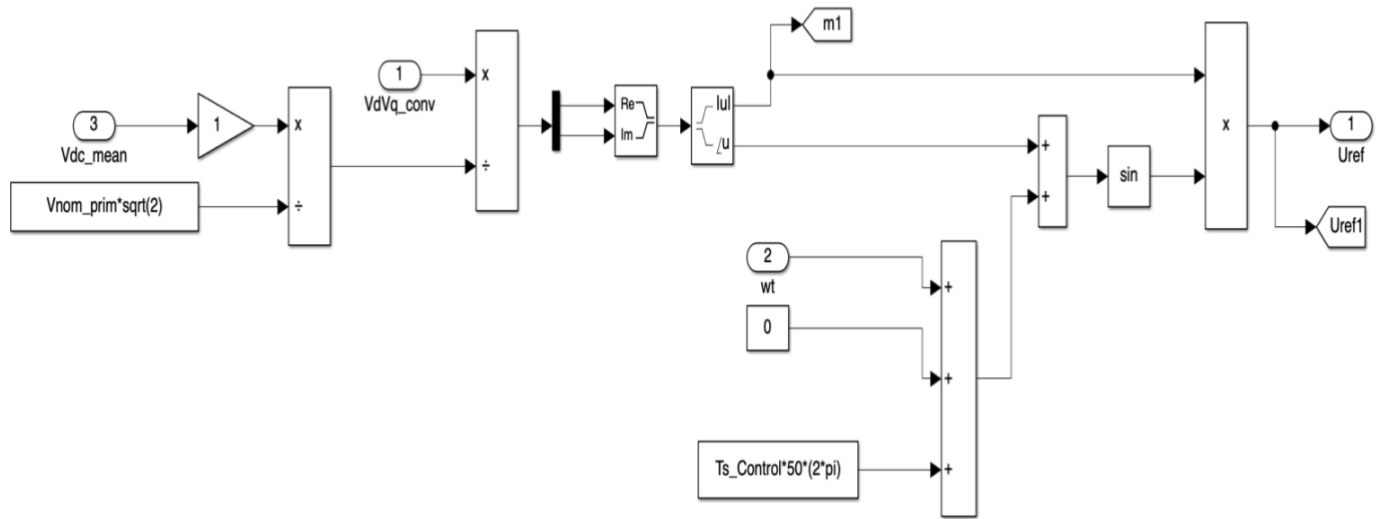


Figure 14 Reference voltage generation subsystem used for PWM modulation.

The generated reference voltage signal is then fed into a PWM generator that produces the pulses required for the switching of the semiconductor switches in the single-phase H-bridge inverter. The PWM generator converts the reference voltage into a pulse signal that is required for the modulation of the voltage from the DC source into a sinusoidal AC voltage that is synchronized with the voltage in the electrical network. The coordination of the three stages of the control system enables the inverter to provide the required power from the photovoltaic–battery system to the electrical network in a

stable manner while ensuring that the inverter is synchronized with the voltage in the electrical network.

3.3.5 Energy Management Strategy

The overall energy management strategy implemented in this study is illustrated in Figure 15. The control logic determines the operating mode of the system by evaluating the available photovoltaic power, the household load demand, and the battery SOC.

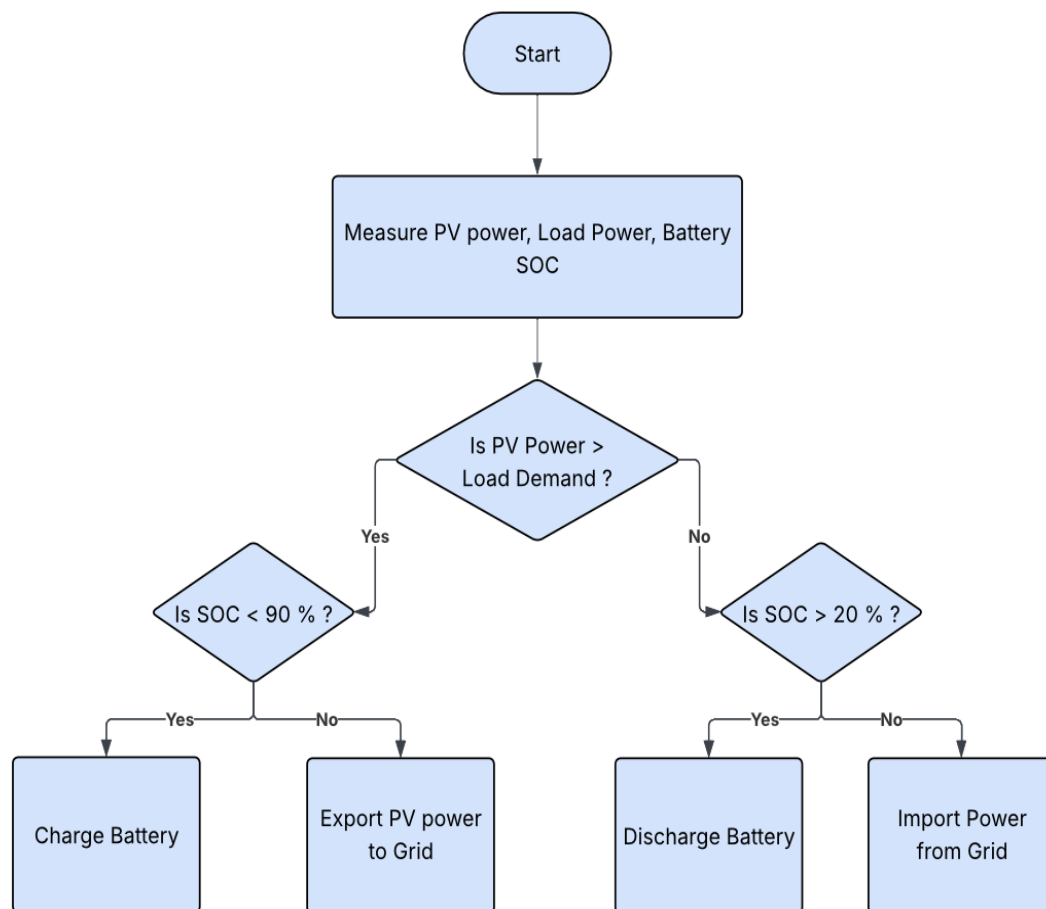


Figure 15 Rule-based energy management strategy for coordinating PV generation, battery storage and grid interaction.

The control logic first compares the available PV power with the household load demand. If the PV generation exceeds the load demand, a surplus of energy is available within the system. In this situation, the controller prioritises storing the excess energy in the battery. Charging is allowed only if the battery SOC remains below the upper limit of 90%, ensuring that the battery operates within safe limits. Once the battery reaches this threshold, any remaining surplus PV power is exported to the grid.

Conversely, when the PV generation is lower than the household demand, the system enters a deficit condition. In this case, the controller evaluates the battery SOC to determine whether the battery can support the load. If the SOC remains above the lower limit of 20%, the battery is allowed to discharge in order to supply the deficit power.

However, if the battery SOC falls below this minimum threshold, further discharge is prevented to protect the battery from deep discharge. Under these conditions, the required power is instead imported from the grid to meet the household demand.

This rule-based strategy therefore enables coordinated power flow between the PV system, battery storage, and the grid while maintaining safe battery operating limits. Rule-based energy management strategies are often preferred for residential PV–battery systems due to their simplicity, low computational requirements, and ease of implementation compared to more complex optimisation techniques (Li and Preindl, 2023). The practical implementation of this decision logic within the simulation environment is presented in Figure 16.

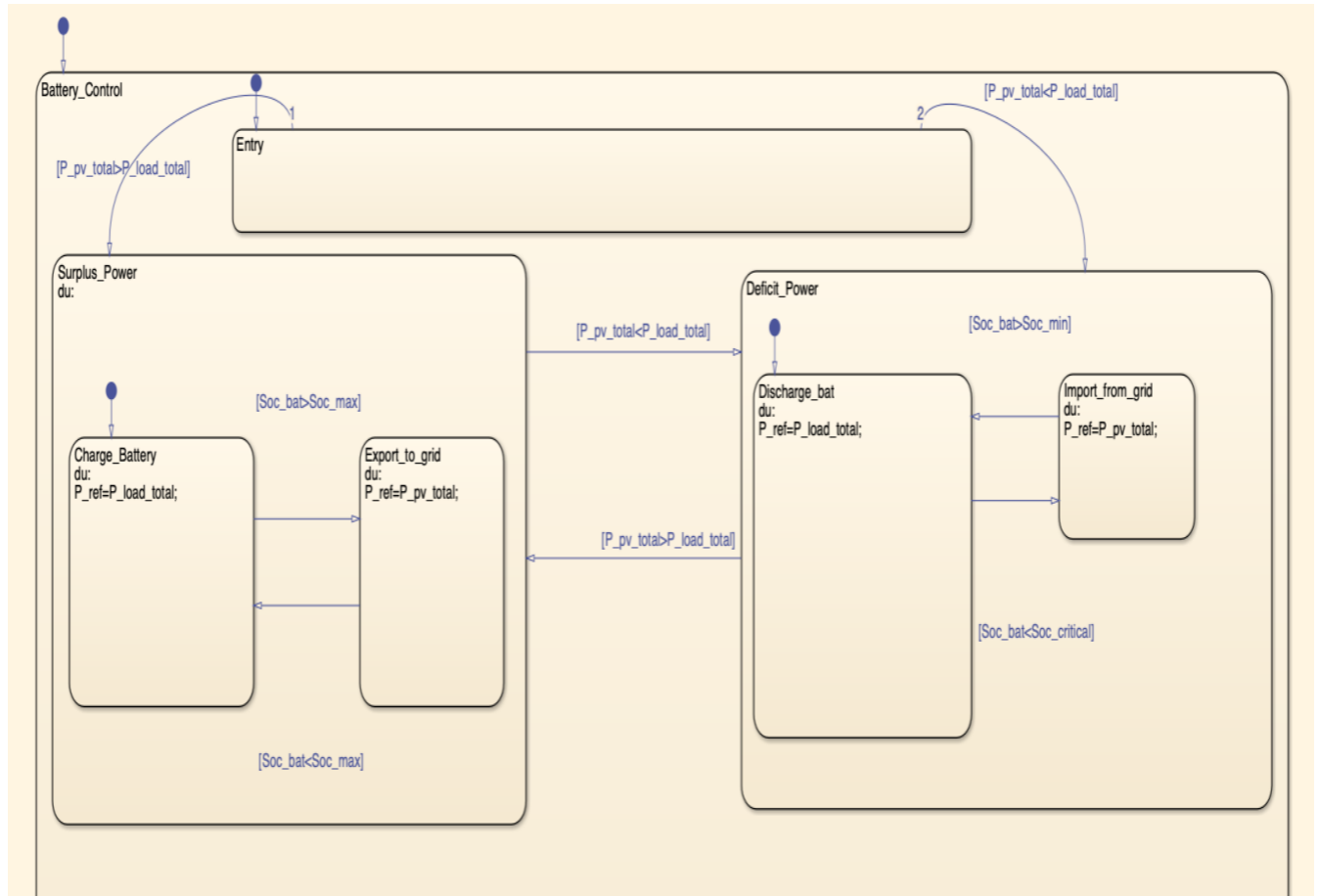


Figure 16 Simulink implementation of the battery energy management control logic.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

4.1 Simulation Setup

The system was simulated in MATLAB/Simulink using a discrete-time model with a stop time of 0.8 s. A power system sampling time of 1×10^{-6} s and a control sampling time of 1×10^{-5} s were used to ensure accurate switching behaviour and stable control performance.

The system operates at a nominal grid voltage of 230 V with a rated power of 4 kW. The DC-link voltage is regulated at 400 V using a PI controller. The battery has a nominal voltage of 96 V and a capacity of 50 Ah, and is connected through a bidirectional converter, allowing both charging and discharging. A constant load was assumed to clearly analyse the effect of PV generation.

The PV array characteristics are shown in Figure 17, presenting the I–V and P–V curves under different irradiance levels. These were used to define the simulation conditions.

Different scenarios were then created by varying PV generation, load demand, and battery SOC to evaluate system performance.

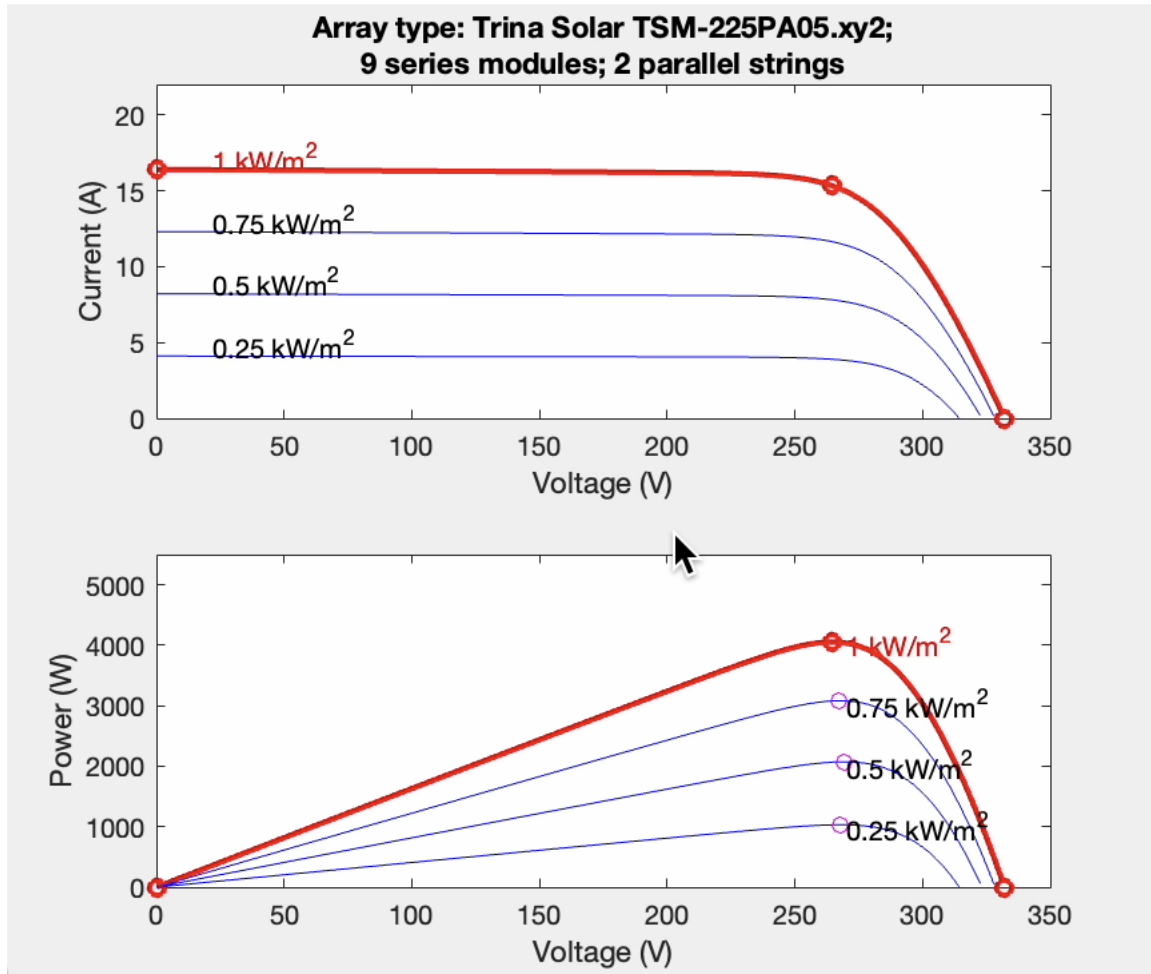


Figure 17 Photovoltaic array I–V and P–V curves at different irradiance levels.

4.2 Energy Management Operating Conditions

To validate the behaviour of the implemented energy management strategy, several operating conditions were simulated. These scenarios represent different combinations of PV generation, load demand, and battery SOC limits.

Table 1 summarises the different operating conditions and the corresponding power flow decisions implemented by the controller.

Table 1 – Operating conditions and corresponding energy management actions in the PV–BESS system.

PV Power vs. Load Power	Battery SOC	Action / Power Flow
PV > Load	< 90%	PV supplies the Load; remaining power charges the Battery.
PV > Load	> 90%	PV supplies the Load; excess power is delivered to the Grid.
PV < Load	> 20%	PV supplies what it can; Battery covers the deficit to power the Load.
PV Not Available	> 20%	Battery supplies all power required by the Load.
PV < Load	< 20% (Option A)	PV supplies what it can; Grid supplies the extra power needed.
PV < Load	< 20% (Option B)	Grid supplies all Load power; Grid also charges the Battery (rate decided by user preference).

4.3. Condition 1 – PV Generation Greater than Load (Battery Charging)

The first scenario corresponds to a condition where PV generation exceeds the load demand while the battery SOC remains below its upper limit. Under this operating condition, the surplus PV power is used to charge the battery. This typically occurs during

daytime periods with high solar irradiance, where generation surpasses household consumption, allowing excess energy to be stored for later use.

The system behaviour under this condition is illustrated through the simulation results shown in Figures 18–20. The corresponding outputs include the PV power, battery SOC, and the power flow at Bus 1 and the grid interface.

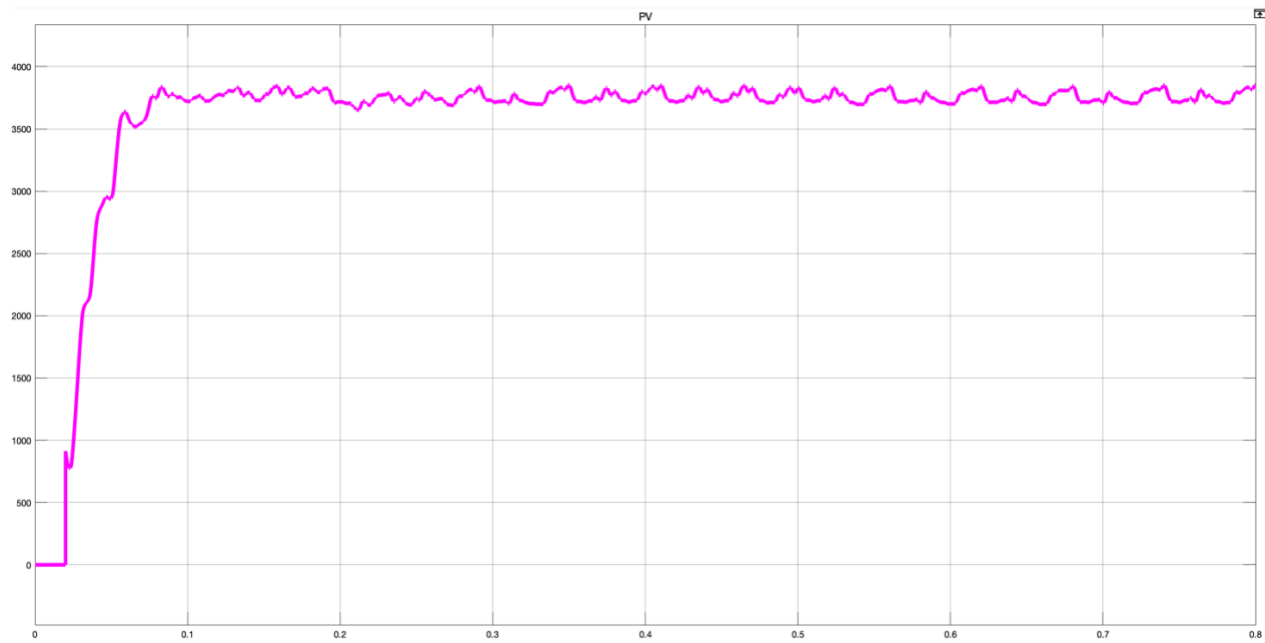


Figure 18 PV power output showing MPPT response and steady-state behaviour.

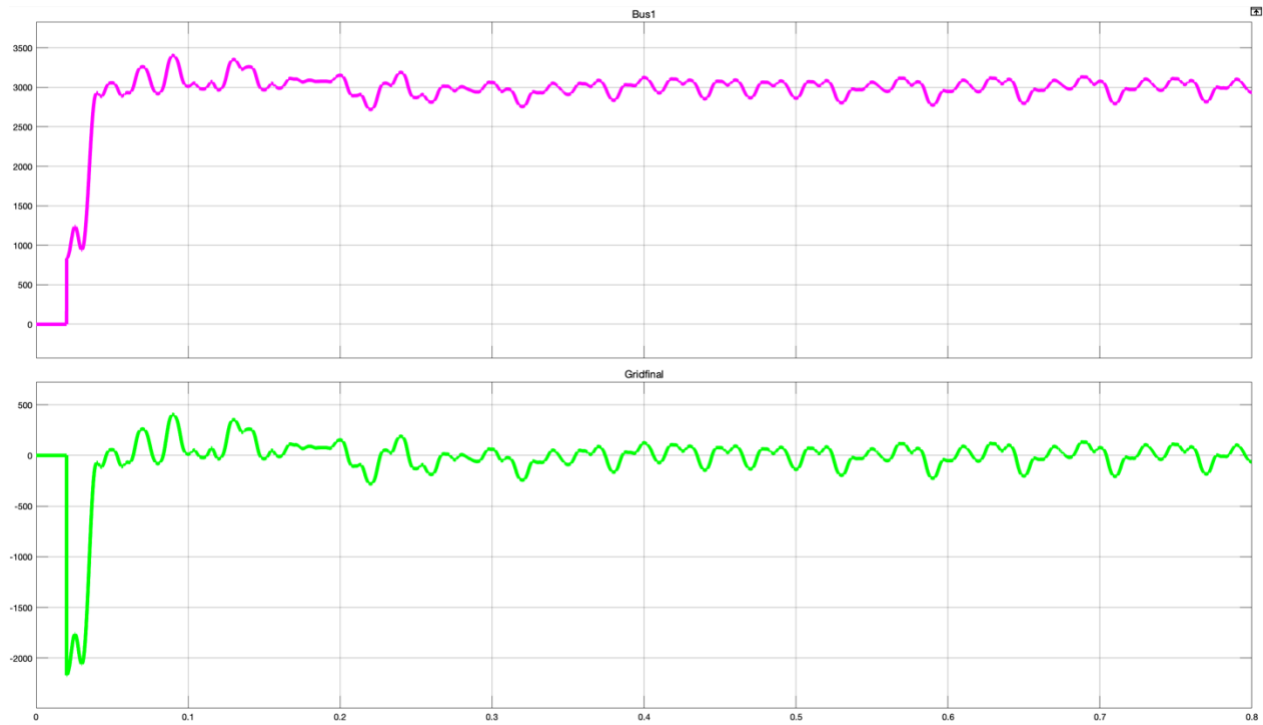


Figure 19 Power at Bus 1 and grid interaction.

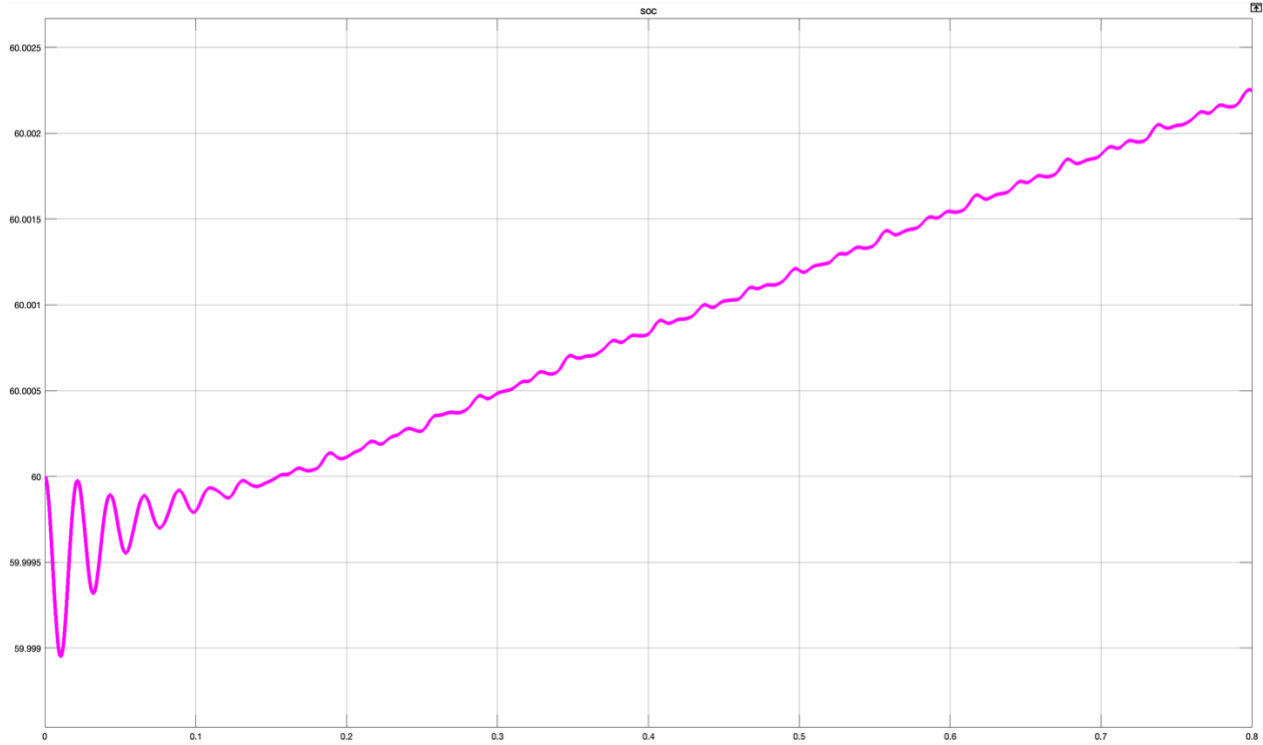


Figure 20 Battery state of charge behaviour.

4.4 Condition 2 – PV Generation Greater than Load with High Battery SOC (Grid Export)

The second operating scenario occurs when the photovoltaic generation remains higher than the load demand, but the battery has already reached its maximum SOC limit. This scenario occurs when solar production is high, and the battery is already fully charged. Since no additional energy can be stored, the excess electricity is exported to the utility grid. The system behaviour under this condition is illustrated through the simulation results presented in Figures 21–23. The corresponding outputs include the PV power, battery SOC, and the power flow at Bus 1 and the grid interface.

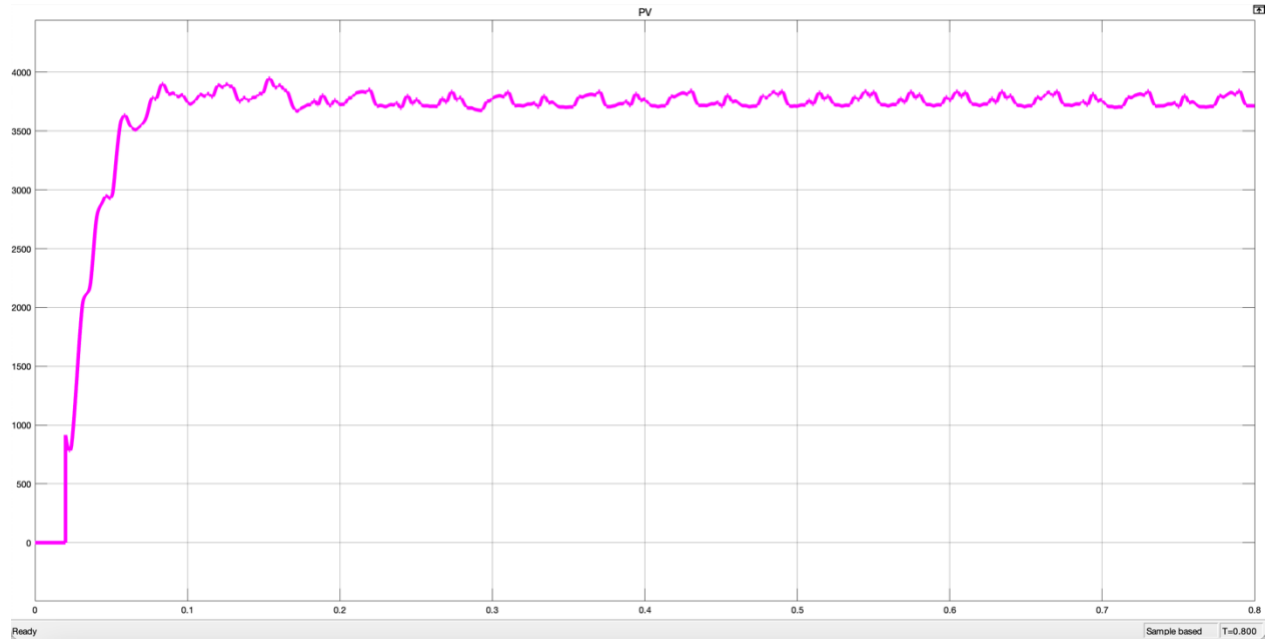


Figure 21 PV power output showing MPPT response and steady-state behaviour.

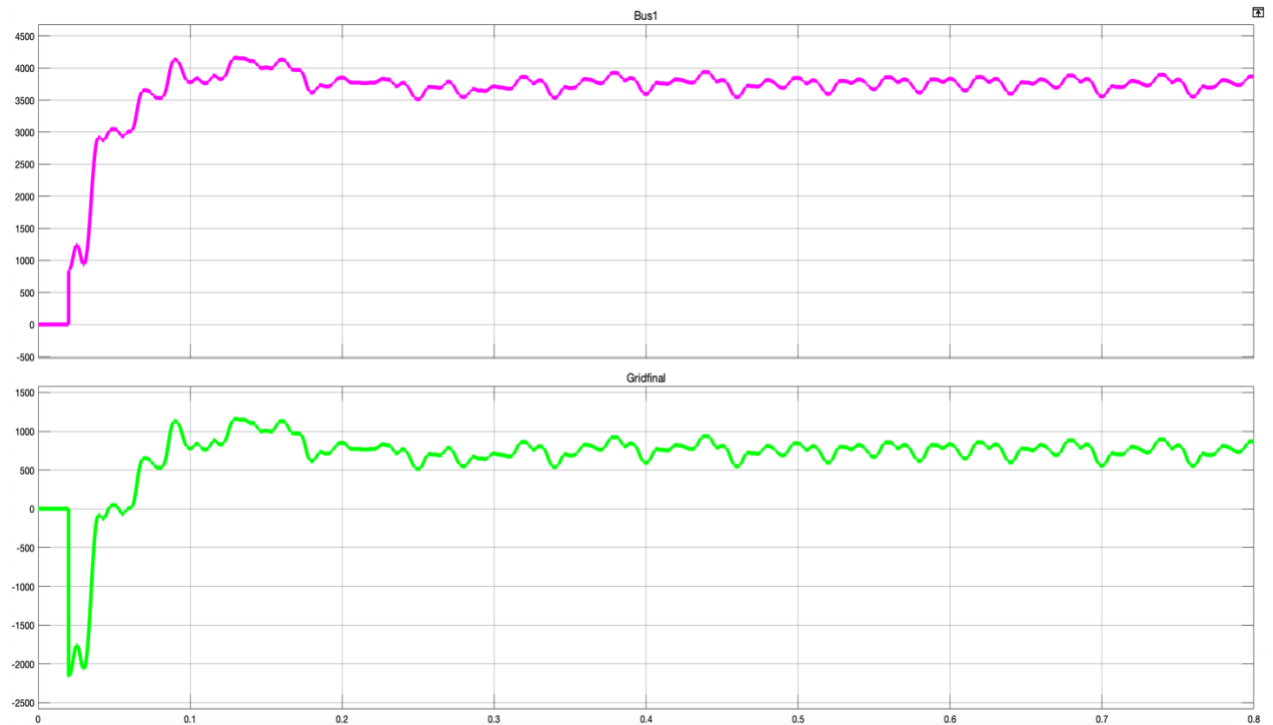


Figure 22 Power at Bus 1 and grid interaction.

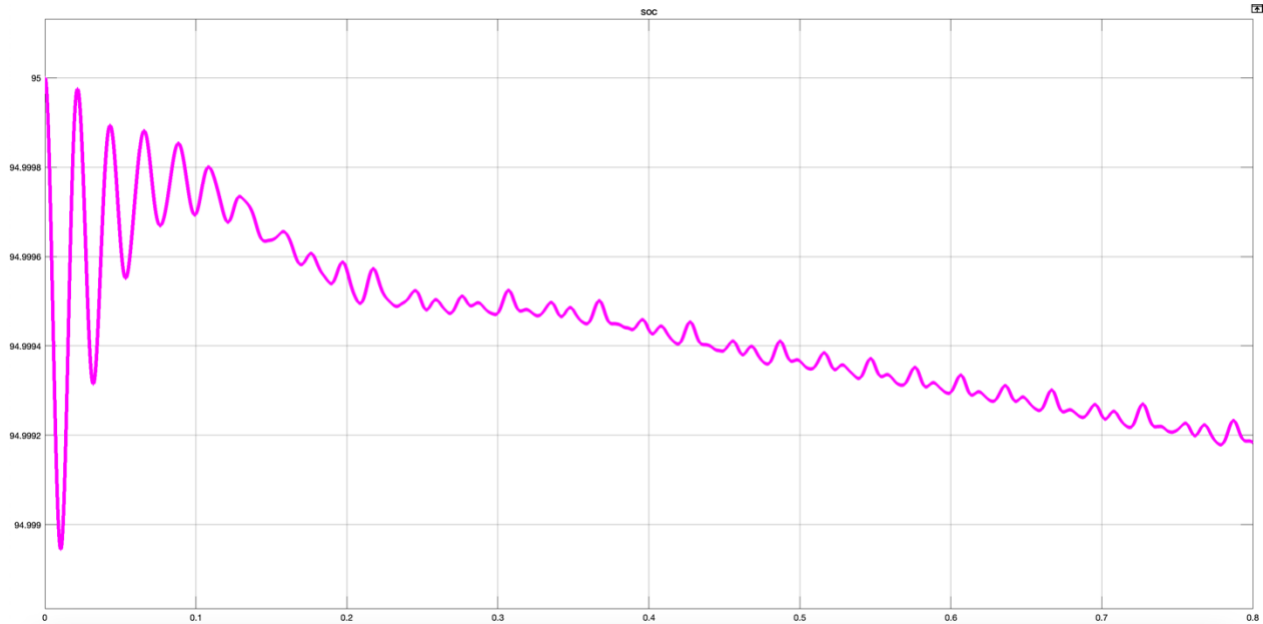


Figure 23 Battery state of charge behaviour.

4.5 Condition 3 – PV Generation Lower than Load (Battery Discharging)

In the third scenario, the photovoltaic generation becomes lower than the load demand. In this situation, the battery is used to supply the deficit power as long as the SOC remains above the minimum threshold. This condition typically occurs during late afternoon or cloudy periods when solar generation decreases while household demand remains high. The battery therefore discharges to supply the additional required energy. The system behaviour under this condition is illustrated through the simulation results presented in Figures 24–26. The corresponding outputs include the PV power, battery SOC, and the power flow at Bus 1 and the grid interface.

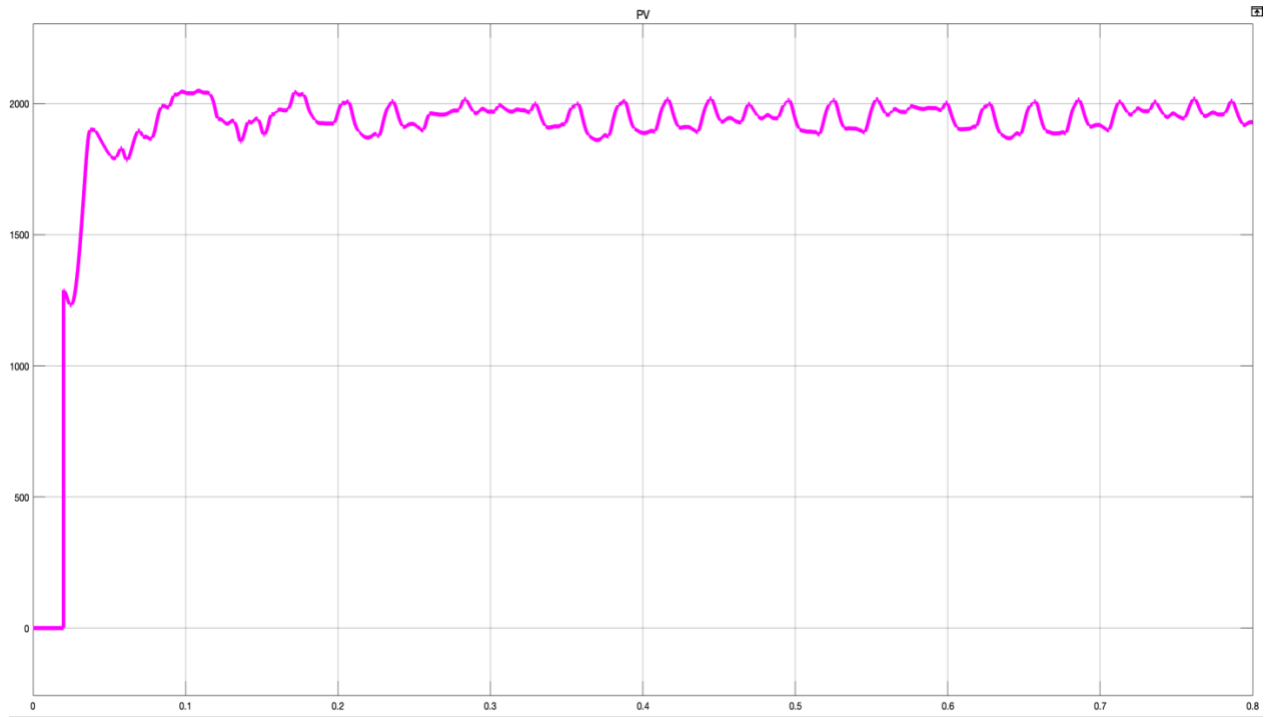


Figure 24 PV power output showing MPPT response and steady-state behaviour.

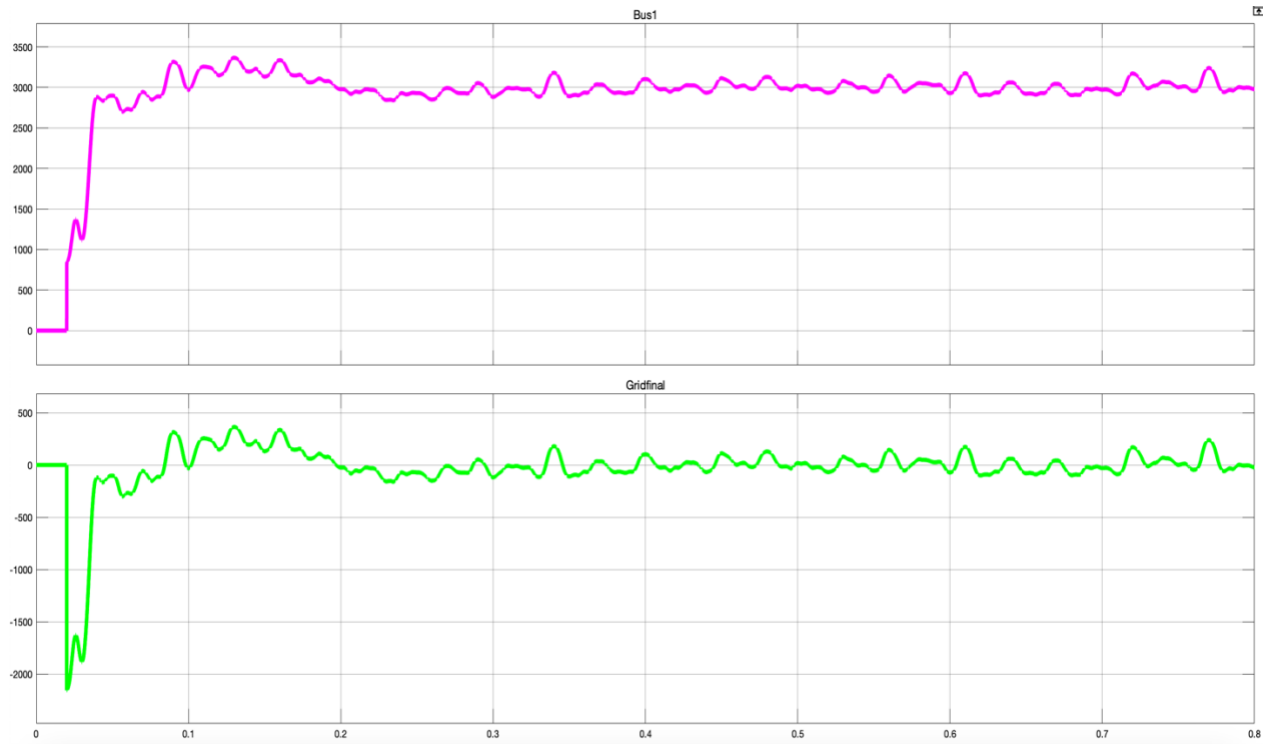


Figure 25 Power at Bus 1 and grid interaction.

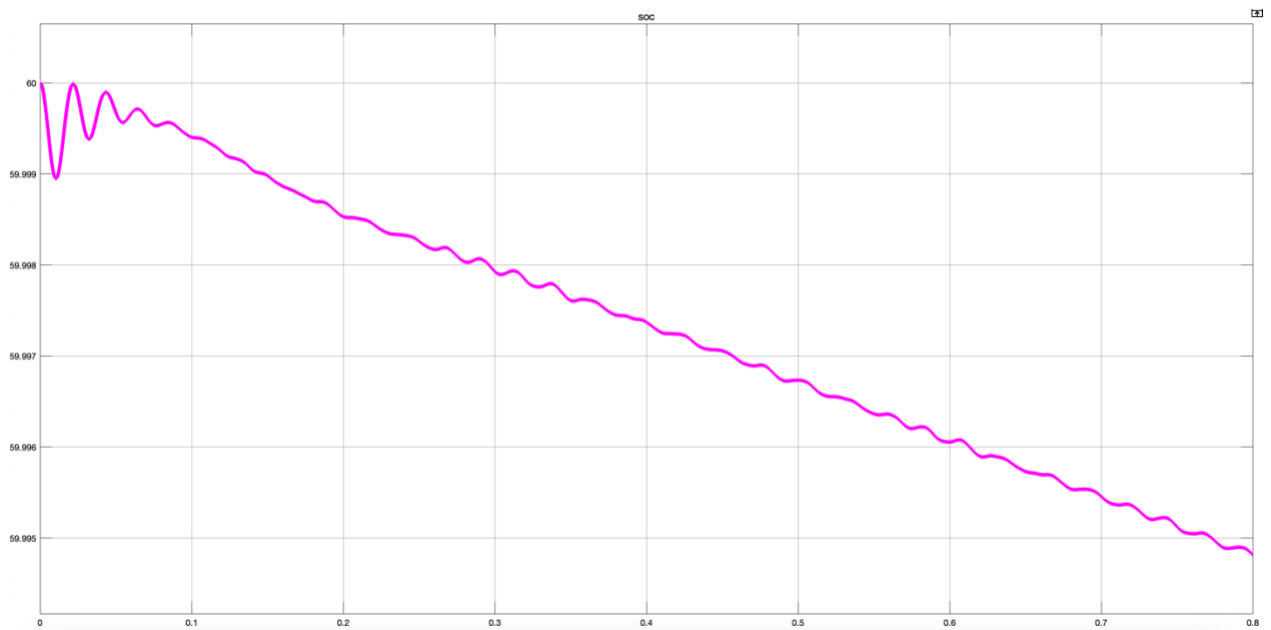


Figure 26 Battery state of charge behaviour.

4.6 Condition 4 – PV Not Available

This situation represents nighttime conditions when the photovoltaic system produces no electricity. The household demand is therefore supplied by the battery until the stored energy is depleted. The system behaviour under this condition is illustrated through the simulation results presented in Figures 27–29. The corresponding outputs include the PV power, battery SOC, and the power flow at Bus 1 and the grid interface.

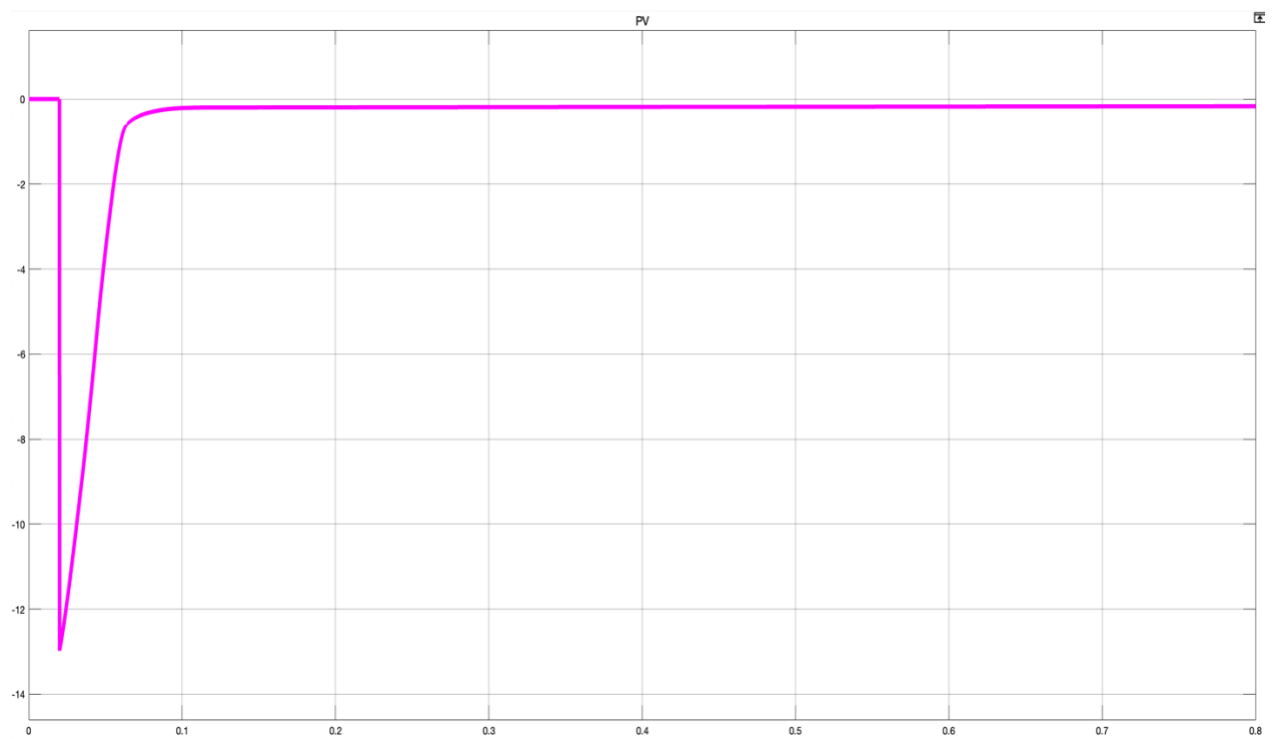


Figure 27 PV power output behaviour.

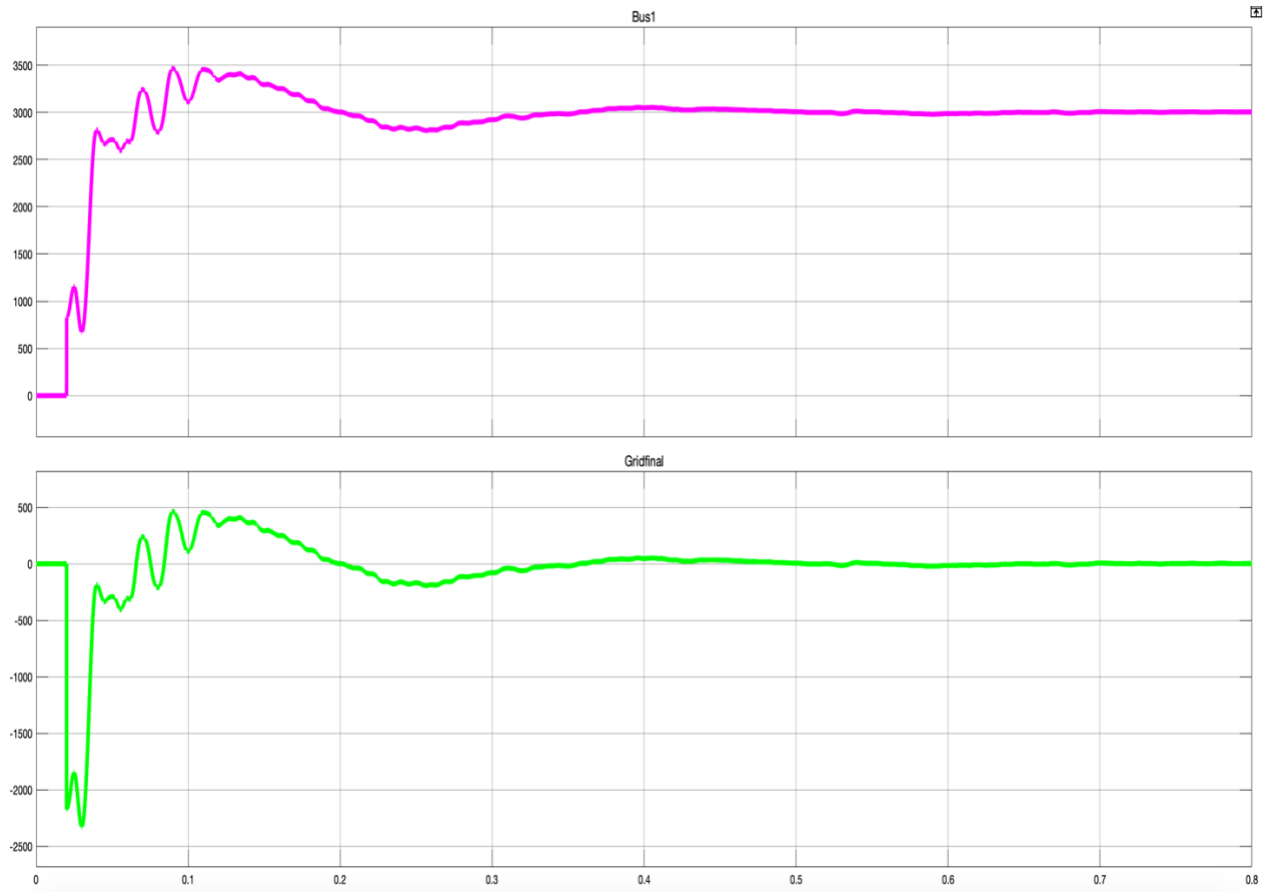


Figure 28 Power at Bus 1 and grid interaction.

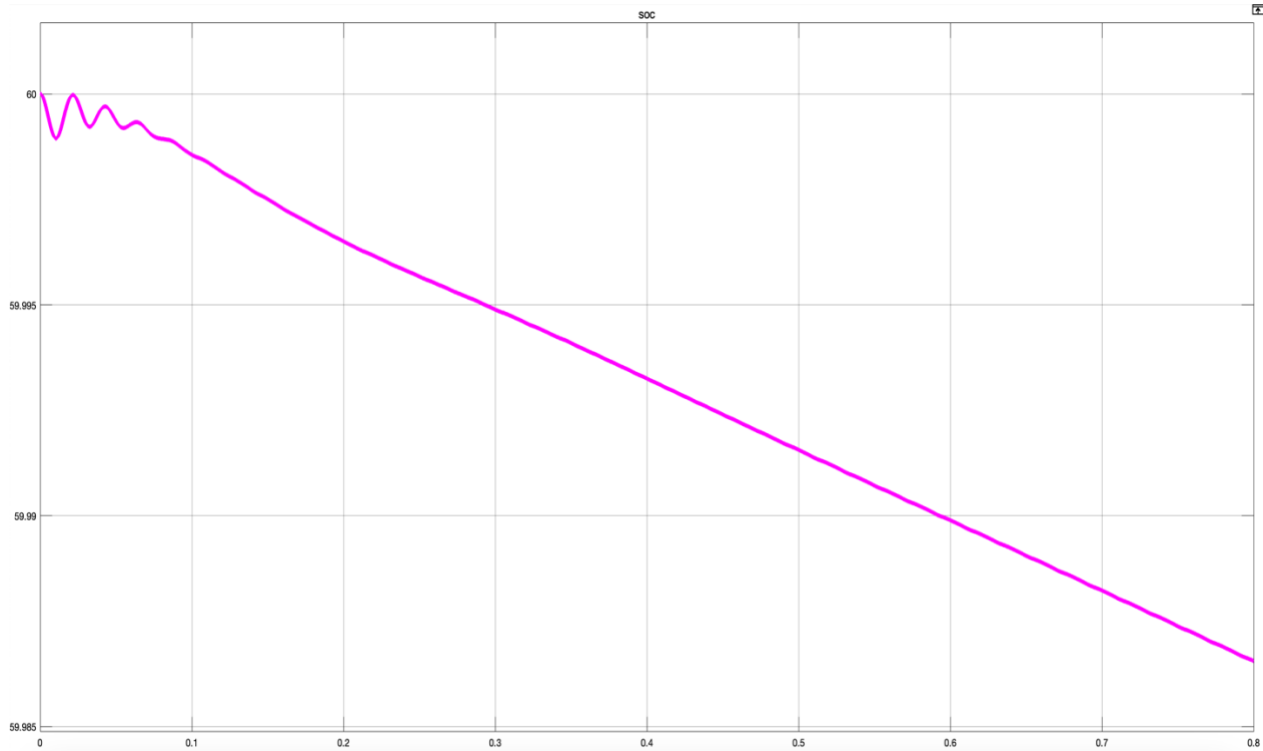


Figure 29 Battery state of charge behaviour.

4.7 Condition 5 – Low Battery SOC (Grid Import)

The final scenario represents the situation where the battery SOC reaches its minimum threshold. The household then relies on electricity imported from the utility grid to meet demand. To protect the battery from deep discharge, the controller prevents further battery discharge once this limit is reached. The household then relies on electricity imported from the utility grid to meet demand. The system behaviour under this condition is illustrated through the simulation results presented in Figures 30–32. The corresponding outputs include the PV power, battery SOC, and the power flow at Bus 1 and the grid interface.

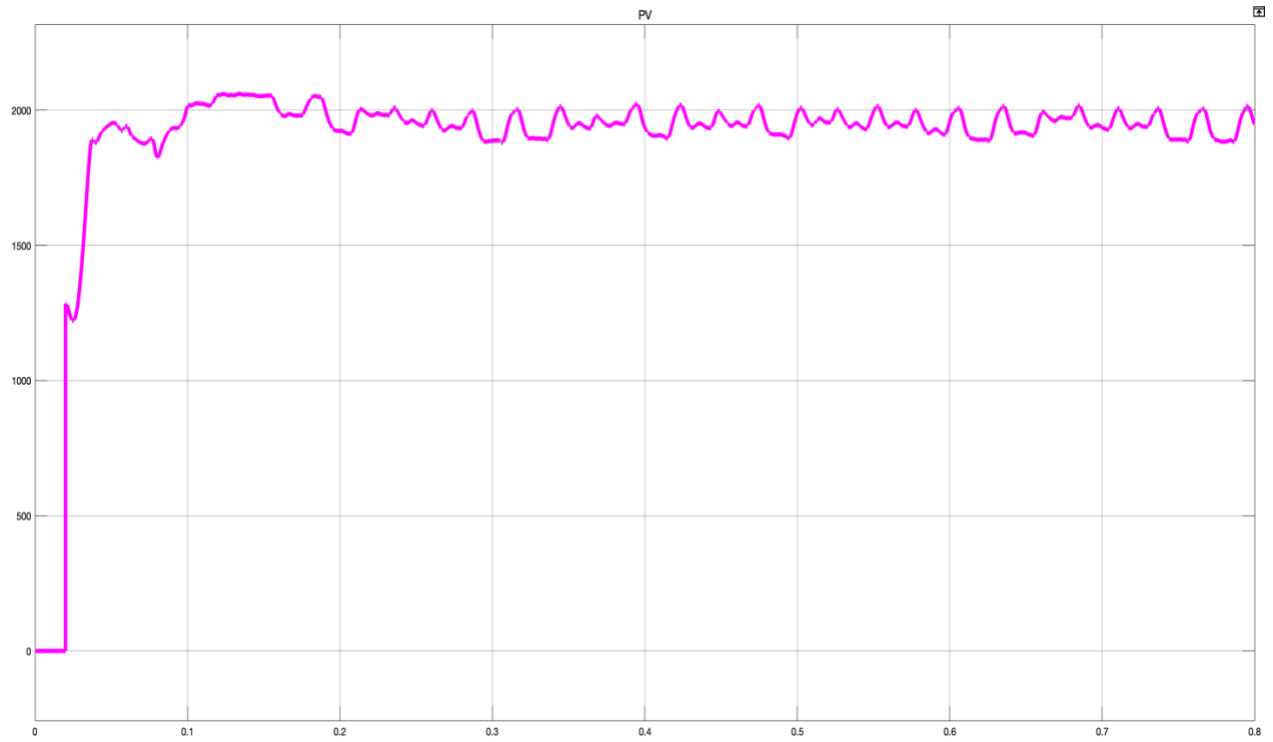


Figure 30 PV power output showing MPPT response and steady-state behaviour.

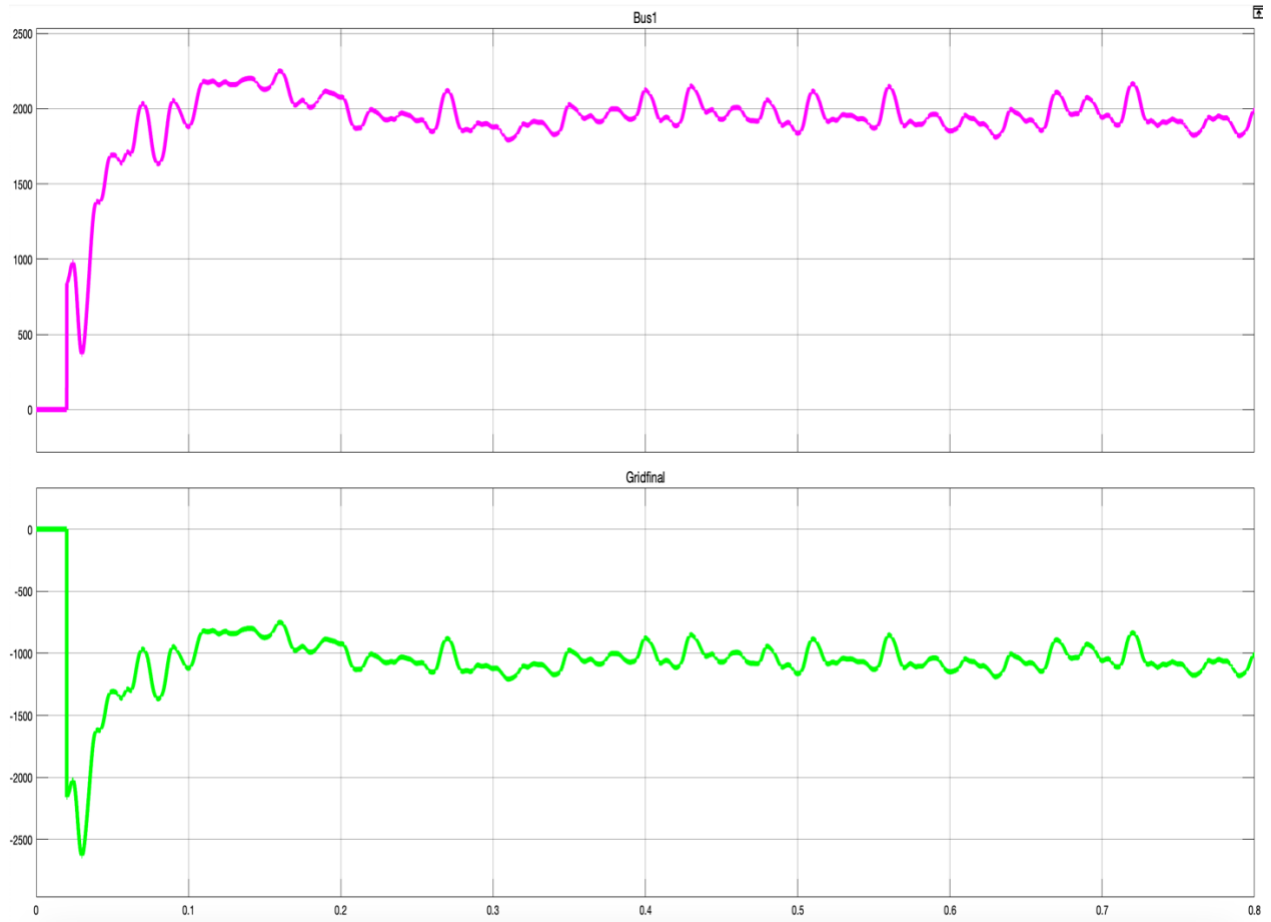


Figure 31 Power at Bus 1 and grid interaction.

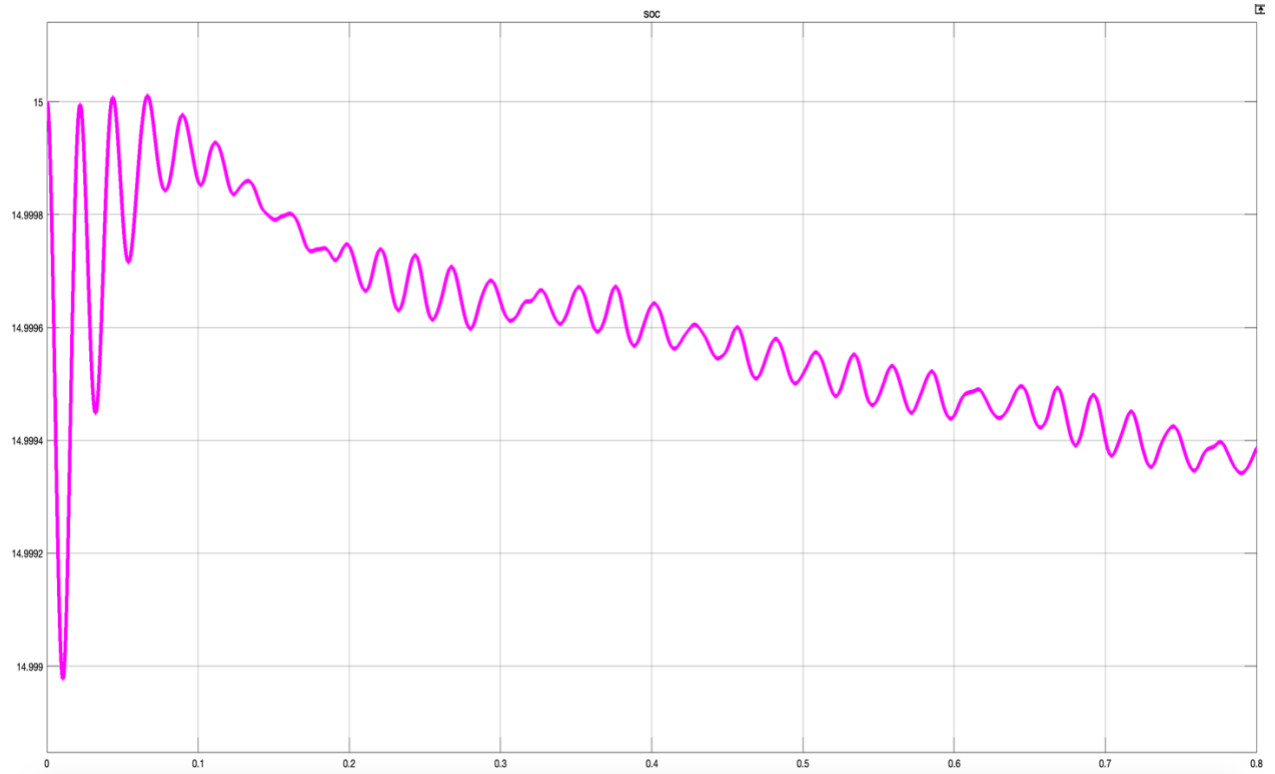


Figure 32 Battery state of charge behaviour.

4.8 Discussion of Results

4.8.1 Operation under High PV Generation and Battery Charging Conditions

Under high irradiance conditions, the PV system generates power close to its maximum capacity, settling around 3.7–3.8 kW after the initial transient period. The small oscillations observed in steady state are expected when using the perturb and observe method, as it continuously adjusts the operating point around the maximum power point. This behaviour confirms that the PV system is operating efficiently and is able to extract almost all the available solar energy under favourable conditions, while maintaining stable converter performance.

At the same time, the battery state of charge increases gradually, which clearly indicates that the excess energy is being stored rather than wasted. Although the increase in SOC is relatively small within the simulation time, the trend is consistent and confirms correct operation. This behaviour reflects the intended energy management logic, where surplus PV generation is prioritised for storage. From a practical perspective, this contributes to better energy utilisation and reduces dependence on the grid.

The load demand remains stable at approximately 3.0 kW, and the grid power remains close to zero after the initial transient. This shows that the system is able to operate in a fully self-sustained mode under these conditions. Overall, this scenario demonstrates effective coordination between generation, storage, and load, validating the performance of the control strategy when PV generation is high.

This behaviour is consistent with findings reported by Deshmukh et al. (2023), where a grid-connected PV–battery system was shown to store excess photovoltaic energy during high generation periods while maintaining stable system operation and reducing reliance on the grid.

4.8.2 Behaviour under High PV Generation with Near-Full Battery State

When the battery approaches its upper SOC limit, the system behaviour begins to adapt due to the reduced storage capacity. The PV output remains stable at around 3.7–3.8 kW, which confirms that the generation side continues to operate efficiently without any noticeable degradation in performance.

Ideally, once the battery reaches a high SOC threshold, it should stop charging completely and all excess energy should be exported to the grid. However, the results show a slight decrease in SOC rather than a perfectly flat profile. This indicates that the battery is still slightly involved in the system dynamics instead of being fully isolated. This behaviour is likely due to the absence of a strict SOC cut-off in the control logic or small control tolerances within the system.

The load demand increases slightly to around 3.8 kW, and the grid power becomes positive, indicating that excess energy is being exported. This confirms that the system is generally behaving as expected in this condition. However, the minor SOC variation highlights a limitation in the control strategy, as the battery is not fully protected at high SOC levels. Despite this, the system still performs efficiently and ensures that excess renewable energy is utilised rather than curtailed.

4.8.3 System Response under Reduced PV Generation with Battery Support

Under reduced irradiance conditions, the PV output decreases to approximately 2.0 kW, which is clearly insufficient to meet the load demand. Despite this reduction, the PV system continues to operate near its maximum available power, as shown by the stable waveform and the small oscillations associated with continuous tracking.

To compensate for the shortfall, the battery begins to discharge, as indicated by the steady decrease in SOC. The SOC curve shows a smooth and continuous decline, which suggests that the battery and its converter are operating in a stable and controlled manner. This confirms that the battery is effectively integrated into the system and is able to support the load when PV generation is limited. This aligns with the behaviour described by Deshmukh et al. (2023), where the battery discharges to compensate for reduced photovoltaic output, ensuring continuous supply to the load without significant grid dependence.

The load demand remains constant at around 3.0 kW, demonstrating that the system maintains reliable supply even under less favourable conditions. The grid power remains close to zero, which means that the battery fully compensates for the deficit without requiring external support. This behaviour highlights the effectiveness of the energy management strategy in prioritising stored energy and maintaining system independence from the grid.

4.8.4 Operation under Zero PV Generation (Battery-Dominant Mode)

When there is no solar generation, the PV output drops to zero, representing conditions such as night-time operation. In this case, the system relies entirely on the battery to meet the load demand, effectively operating in a battery-dominant mode.

The SOC decreases steadily over time, indicating controlled and continuous battery discharge. The smooth profile of the SOC curve shows that the system does not experience instability or abrupt switching, which is important for maintaining reliable operation. This suggests that the control strategy is able to handle this transition effectively.

The load demand remains at approximately 3.0 kW, confirming that the battery is capable of sustaining the system on its own within the simulation timeframe. Although there is a brief grid contribution during the initial transient period, it quickly disappears and becomes negligible in steady state. Overall, this scenario demonstrates that the system can maintain stable operation even in the complete absence of PV generation.

4.8.5 Behaviour under Low SOC Conditions with Grid Support

When the battery approaches its lower SOC limit, the system is expected to reduce or stop battery discharge in order to protect the battery. In this condition, the PV output remains around 2.0 kW, which is still not sufficient to meet the load demand.

However, the results show that the SOC continues to decrease slightly, which means that the battery is still contributing even near its minimum threshold. Ideally, the battery should stop discharging completely at this point, allowing the grid to supply the remaining demand. The observed behaviour suggests that the SOC constraint is not strictly enforced within the control strategy.

The grid power becomes negative and stabilises around -1.0 kW, confirming that energy is being imported from the grid. However, since the battery is still discharging slightly, both the battery and grid are sharing the load. This indicates that the system does not fully switch to grid-only operation when the SOC limit is reached. While this highlights a limitation in the control design, the system still maintains stable operation and continues to meet the load demand without interruption.

4.8.6 Overall System Performance Analysis

Across all operating conditions, the system demonstrates stable and consistent performance in maintaining load supply. One of the most important observations is that the load demand remains close to its nominal value in all scenarios, which confirms the reliability of the system.

The PV system consistently operates near its maximum available power under different irradiance levels, showing that energy extraction is effective. The small oscillations observed in the PV output are expected and confirm that the system is continuously tracking the optimal operating point.

The battery plays a key role in balancing the system by storing excess energy and supplying power when needed. The SOC profiles clearly reflect different modes of operation, including charging, discharging, and operation near limits. However, slight deviations from ideal behaviour near SOC boundaries indicate that the control strategy could be improved further.

Grid interaction is generally minimised, with the system prioritising local generation and storage. The grid is only used when necessary, either to export excess energy or to support the system when both PV and battery are insufficient. Overall, the results demonstrate that the energy management strategy is effective, adaptable, and capable of maintaining stable operation under a wide range of conditions.

CHAPTER FIVE

CONCLUSIONS AND FUTURE WORK

5.1 Key findings

5.1.1 Reliable PV System Modelling and Operation

The MATLAB/Simulink model developed in this project was able to successfully integrate the PV system, power converters, battery storage, and grid connection into a single working system. The model performed consistently across all tested conditions, including changes in irradiance, load demand, and battery state of charge. This shows that the system is not only functional but also stable under different operating scenarios. Overall, the model provides a realistic representation of a residential PV-battery system and confirms that the overall system design works as intended.

5.1.2 Effective Maximum Power Extraction

The use of the Perturb and Observe MPPT algorithm allowed the PV system to operate close to its maximum power point under different irradiance levels. The output power followed the expected behaviour, with only small oscillations around the steady value, which is normal for this type of controller. This means that the system is able to make

good use of the available solar energy rather than wasting it. As a result, the MPPT implementation plays an important role in improving the overall efficiency of the system.

5.1.3 Stable Grid-Connected Power Delivery

The grid-connected inverter operated smoothly and allowed power to be exchanged between the system and the grid without any major instability. The results showed that the power flow remained stable and well controlled, even when the system conditions changed. The DC-link voltage was also maintained within a suitable range, which is essential for proper inverter operation. This indicates that the inverter and its control are working correctly and are capable of supporting reliable grid-connected operation.

5.1.4 Functional Battery Model with SOC Behaviour

The battery model worked effectively and responded well to different operating conditions. The SOC changed as expected, increasing when there was excess PV generation and decreasing when the battery was used to support the load. This behaviour confirms that the battery is properly integrated into the system and plays its role in balancing energy supply and demand. The SOC monitoring also helps ensure that the battery operates within safe limits, which is important for realistic system operation.

5.1.5 Effective Energy Management Strategy

The energy management strategy was able to control how power flows between the PV system, the battery, and the grid in a logical and effective way. The system prioritised the use of solar energy, stored excess energy in the battery when possible, and only relied

on the grid when necessary. This shows that the strategy works well in managing different situations and helps reduce dependence on the grid. Overall, it demonstrates that a simple control approach can still achieve good performance in a residential PV-battery system.

5.2 Conclusions

The developed PV–battery system produced consistent and reliable results across all simulated operating conditions. The system demonstrated effective performance in extracting maximum available power from the PV source, with stable operation observed under both high and low irradiance levels. The energy management strategy enabled appropriate coordination between the PV system, battery, and grid, ensuring that power was delivered to the load without interruption.

The results confirmed that the system adapts well to changing operating conditions. Under high PV generation, excess energy was successfully utilised, while under reduced or zero generation, the battery and grid supported the load as required. The power profiles remained stable throughout, indicating proper system operation and control.

Overall, the study can be considered successful as the system operated as intended and maintained reliable performance across all scenarios. The integration of PV generation, battery storage, and grid interaction proved effective for managing residential energy demand in a consistent and practical manner.

5.3 Future Work

Further work beyond the scope of this project may focus on improving the system performance, control strategy, and practical implementation. Possible future developments include:

1. Advanced Energy Management Control Strategies:

Future work can focus on improving the control algorithm to ensure stricter enforcement of SOC limits and more optimal power flow between the PV system, battery, and grid. More advanced control techniques, such as adaptive or predictive control, could enhance system efficiency and reliability.

2. Improved MPPT Techniques:

Although the P&O method performed effectively, it introduced small steady-state oscillations. Future work may involve implementing more advanced MPPT algorithms, such as Incremental Conductance or intelligent optimisation methods, to improve tracking accuracy and reduce power losses.

3. Enhanced Battery Modelling:

The battery model used in this study can be extended to include additional factors such as temperature effects, degradation, and ageing. This would provide a more realistic representation of long-term battery performance and improve the accuracy of system simulations.

4. Real-Time and Hardware Implementation:

The current work is limited to simulation. Future studies could include real-time

implementation or hardware-in-the-loop testing to validate the system under practical operating conditions and assess its real-world performance.

5. Extended Economic Analysis:

Further work may involve a more detailed economic evaluation, including real UK electricity tariffs, installation costs, and long-term savings. This would provide a clearer understanding of the financial benefits and feasibility of the proposed system.

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Appendix A: Gantt Chart



Appendix B: Other information

```
function D = Pand0( V, I)

% MPPT controller based on the Perturb & Observe algorithm.

% D output = Reference for DC link voltage (Vdc_ref)
%
% Enabled input = 1 to enable the MPPT controller
% V input = PV array terminal voltage (V)
% I input = PV array current (A)
%
% Param input:
Dinit = 260; %Initial value for Vdc_ref
Dmax = 290; %Maximum value for Vdc_ref
Dmin = 230; %Minimum value for Vdc_ref
deltaD = 0.03; %Increment value used to increase/decrease Vdc_ref
%

persistent Vold Pold Dold;

dataType = 'double';

if isempty(Vold)
    Vold=0;
    Pold=0;
    Dold=Dinit;
end
P= V*I;
dV= V - Vold;
dP= P - Pold;

if dP ~= 0
    if dP < 0
        if dV < 0
            D = Dold + deltaD;
        else
            D = Dold - deltaD;
        end
    else
        if dV < 0
            D = Dold - deltaD;
        else
            D = Dold + deltaD;
        end
    end
else D=Dold;
end

if D >= Dmax | D<= Dmin
    D=Dold;
end

Dold=D;
Vold=V;
Pold=P;
```

Figure 33 MATLAB implementation of the Perturb and Observe (P&O) MPPT controller.

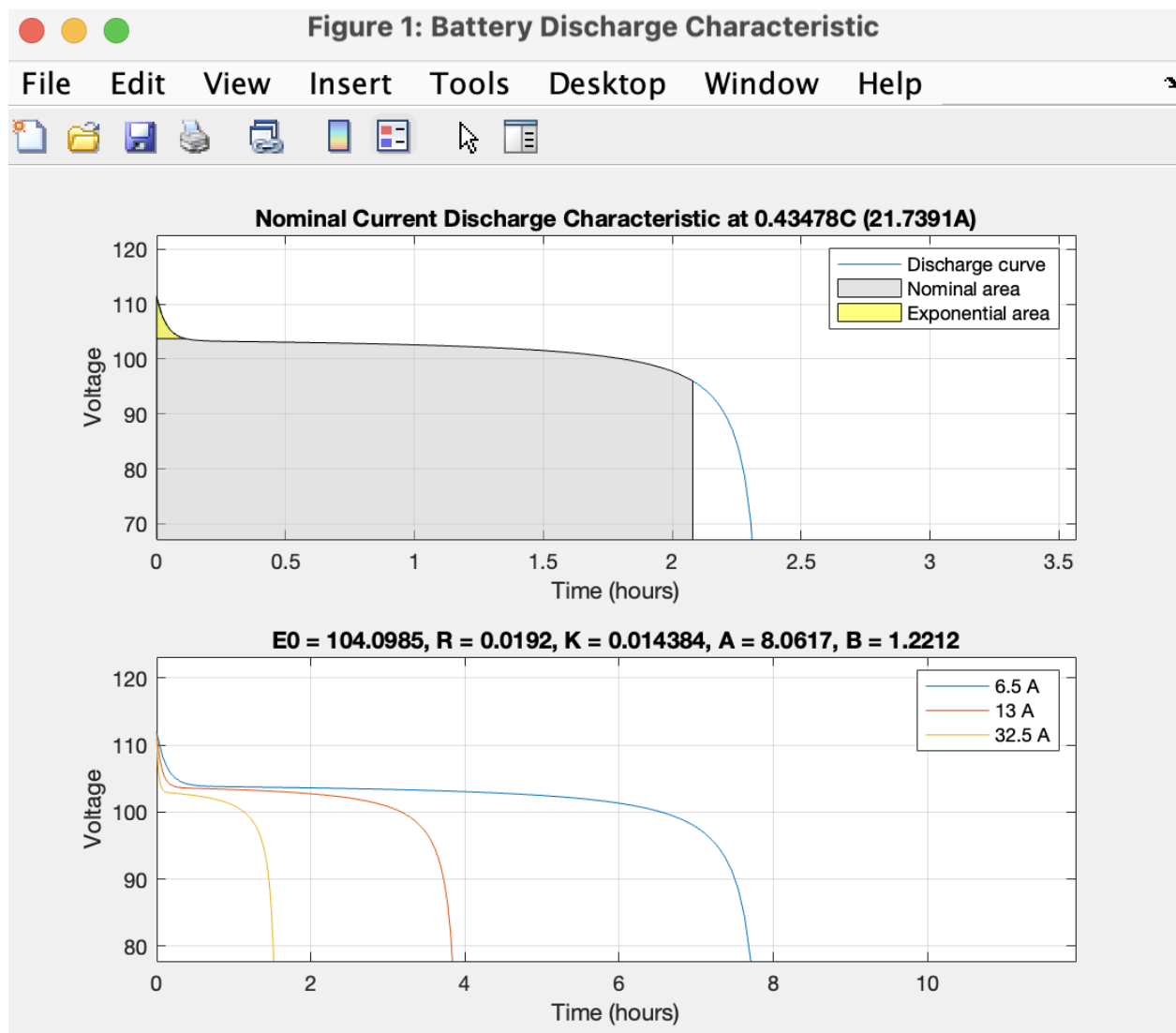
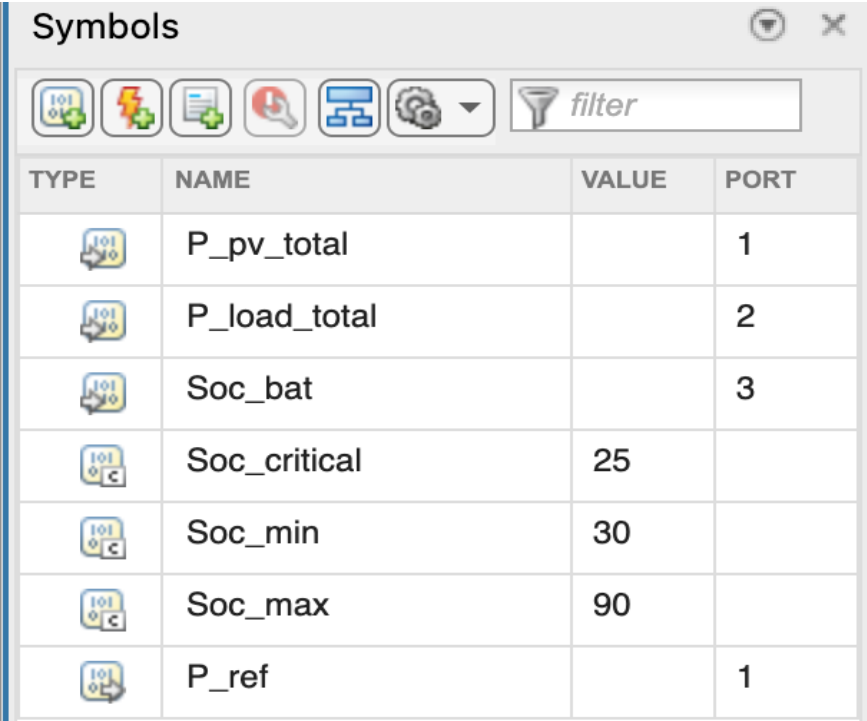


Figure 34 Battery discharge characteristics and model parameters used for simulation, showing voltage response under different discharge currents.



TYPE	NAME	VALUE	PORT
	P_pv_total		1
	P_load_total		2
	Soc_bat		3
	Soc_critical	25	
	Soc_min	30	
	Soc_max	90	
	P_ref		1

Figure 35 Key control variables and SOC threshold parameters used in the energy management strategy implementation.